

An optimisation approach for supply chain procurement transport operational planning with 3D full-truck loading constraints and returnable transport items

This paper proposes a new mixed integer linear mathematical programming (MILP) model, dubbed 3D-SCPTOP, for procurement transport planning in manufacturing supply chains. The model effectively integrates procurement and transport decisions while accounting for three-dimensional full-truck loading constraints and managing returnable transport items (RTIs). Thus, we optimise the material to procure for an assembly manufacturing site, RTIs closed-loop management, and the involved transport decisions, including the truck loading configuration. Procurement transport decisions are based on the demand of assemblers, the supplier’s inventory levels, and the dimensions of the truck / RTIs. The optimisation model aims to minimise transportation and inventory costs. Because the loading subproblem is NP-hard and is replicated once per loading configuration, the compact model is intractable at industrial sizes. We therefore reformulate it by Dantzig–Wolfe decomposition, taking the complete loaded truck as the unit of decision: a column is a fully specified, geometrically realisable truck load, so every placement, height, crush-limit, payload and full-truckload constraint holds inside it by construction, and the per-stack crush cap is evaluated on the actual product mix rather than on a worst-case bound. Columns are priced on demand by a constraint programming oracle and integrality is reached by branch and price, yielding MATH-3D-SCPTOP. Branching is on aggregates — dispatch totals per direction and period, then delivery totals — rather than on the master’s own variables, which a Dantzig–Wolfe master answers merely by shifting fractionality onto a near-identical column; the search keeps both children, and re-prices at every node, so the loads that make an integer plan cheap are generated after the branching decisions rather than assumed available beforehand. We report a lower bound only when the pricing oracle has proved that no column prices out, and none otherwise. Every plan the method returns is re-verified truck by truck, independently of the solver that produced it. The approach was validated with a case study in an automotive seating first-tier supplier, where it improves on the company’s spreadsheet-based procedure in both transport and inventory terms. A comparison against a benchmark model that represents truck capacity volumetrically shows why the loading geometry cannot be abstracted away: the volumetric model returns cheaper plans whose truck loads cannot physically be loaded.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

1 Introduction

Optimizing the use of resources in manufacturing supply chains is crucial for boosting economic growth, improving social well-being, and minimizing environmental impact. Integrating transport, inventory, and production processes is a fundamental characteristic of supply chains (Stadtler, Kilger, and Meyr 2015) because joint decisions will improve supply chain performance (Díaz-Madroño, Mula, and Peidro 2017).

Procurement refers to the process of acquiring goods and services in a supply chain (Chopra and Meindl 2015). One of the main goals of procurement is to obtain raw materials, parts or finished goods for a factory or distribution centre from a supplier. As mentioned in (Fleischmann, Kopfer, and Sürie 2015), the responsibility for transport planning usually lies with the supplier. However, sometimes the manufacturer makes transport decisions of its supplies in, for example, the automotive industry. Furthermore, an essential asset in the automotive supply chain is not considered in traditional material requirement planning (MRP): returnable containers or returnable transport items (RTI), which are used as tertiary packaging. RTIs enable the storage and transportation of RM and FG through the supply chain. Usually, the manufacturing plant

supplies empty RTIs to its supplier, while the supplier delivers full RTIs with the FG or RM. Therefore, during the procurement process, transport and inventory decisions should be integrated, including RTI management. Here, we focus on optimising the operational decisions made during procurement transport planning.

According to (Díaz-Madroño, Peidro, and Mula 2014), using algorithm-based methods for procurement transport planning is more effective than relying on manual approaches based on planners’ intuition. Manual planning can be short-sighted and lead to less efficient outcomes. This becomes even more significant when optimizing procurement transport tasks that aim to reduce the number of trucks and inventory levels, especially under 3D loading constraints where the precise placement of containers within trucks must also be determined (cite). Furthermore, the capacity constraints used in the scientific literature on procurement transport planning are based mainly on freight volume or the total number of loaded product units (Hashemi-Amiri, Ghorbani, and Ji 2023) and (Díaz-Madroño, Peidro, and Mula 2014). The current research seeks to address this gap by introducing a new mixed-integer linear programming (MILP) model. This model uniquely combines procurement transportation with considerations for 3D loading, inventory management, and RTI management. Hereafter, we refer to this model formulation as 3D supply chain procurement transport operational planning (3D-SCPTOP). Additionally, a matheuristic procedure (MATH-3D-SCPTOP) is researched.

In this paper, we present four main contributions: (i) formulating a novel MILP model, 3D-SCPTOP, that integrates procurement transport, RTI management and inventory decisions with stack-based full-truck loading constraints; (ii) reformulating it by Dantzig–Wolfe decomposition over geometrically realisable truck loads, and solving it by branch and price with a constraint programming pricing oracle, giving MATH-3D-SCPTOP; (iii) showing that a volumetric representation of truck capacity, as used in the procurement transport literature, yields plans that are cheaper on paper but cannot be loaded in practice; (iv) validating and testing the proposal with a case study at a first-tier supplier of the automotive sector.

Readers are referred to the typology by (Wäscher, Haußner, and Schumann 2007) for a global view of container loading problems, where the different dimensions of the loading problem are described. Furthermore, (Silva, Toffolo, and Wauters 2019) summarise the exact methods used in the literature to solve container loading problems, and (X. Zhao et al. 2016) explore other solution approaches in the literature. Readers are also referred to the literature reviews of (Mosca, Vidyarthi, and Satir 2019) and (Díaz-Madroño, Peidro, and Mula 2015), which focus more on integrating inventory decisions and transport. Here, we jointly address container loading, procurement transport and inventory decisions. [Furthermore, we refer readers \(\) for insights about the packaging methods used in supply chains.](#)

The paper is organised as follows. Section 2 provides a literature review related to the problem under study. Section 3 presents the problem description. Section 4 describes the 3D-SCPTOP MILP formulation. Section 5 develops the branch and price solution approach, MATH-3D-SCPTOP, together with its pricing subproblem and bounding arguments. Section 6 reports the computational experiments and Section 7 the case study. Finally, Section 8 concludes and puts forward further research lines.

2 Literature Review

The goal of this research is to integrate procurement transport operational planning decisions with 3D loading restrictions and RTI packaging. Based on the authors’ research, no previous work has been identified that meets this criterion. Therefore, three independent, but sometimes overlapping, knowledge bodies have been reviewed: procurement transport and inventory planning, container loading problems and RTI literature.

2.1 Procurement transport and inventory planning

Models that integrate transport and inventory costs during the procurement process are widely studied in the literature. As noted in (Langley et al. 2020), transportation facilitates the movement of raw materials and finished products between suppliers, warehouses, manufacturers, and retailers. Therefore, transport processes play a crucial role in supply chains. (Fleischmann, Kopfer, and Sürie 2015) distinguish between two main types of transportation within a supply chain: “procurement transport,” which involves moving materials from suppliers to manufacturing sites, and the “distribution of goods,” which concerns delivering finished products from factories to customers. This article focuses specifically on addressing the challenges related to procurement transportation.

There are two main dimensions of procurement transport problems according to the supply chain configuration and the applied transport method. Regarding supply chain configurations (Mosca, Vidyarthi, and Satir 2019): (i) single source to a single destination (SS); (ii) multiple sources to a single destination

(MS); (iii) single source to multiple destinations (SM); (iv) multiple sources to multiple destinations (MM). Source components determine the number of supplier locations that are taken into account, while destination components denote the number of supplied manufacturers. Additionally, different transport methods are identified in (Díaz-Madroño, Peidro, and Mula 2014): milk run, less-than-truckload (LTL), full-truckload (FTL). We also analyze the formulation of the transportation capacity constraints in the scientific literature.

(Berman and Wang 2006) address the problem of optimising distribution strategies to deliver products from suppliers to plants, whose aim is to minimise transport, pipeline and plant inventory costs by using cross docks. A mixed integer non-linear programming (MINLP) formulation, a Lagrangian relaxation heuristic, a greedy heuristic, and a branch-and-bound algorithm are proposed to model and solve the problem.

(Sancak and Salman 2011) present a MILP model to optimise procurement transport decisions in a SS supply chain configuration using LTL and FTL transport methods. The objective is to minimise transport and inventory holding costs, while fulfilling production plan requirements on a multiperiod planning horizon. The study explores the possibility of delaying LTL shipments to the subsequent time period, and utilising safety stocks whenever necessary, in a stochastic environment by employing data from a bus manufacturer. The findings demonstrate that delaying shipments leads to reduced holding and transport costs with a minimal stockout risk.

(Schöneberg, Koberstein, and Suhl 2010) examine the concept of delivery profiles, which define the delivery frequency at a fixed rate, offering structured planning, minimizing management effort, and improving coordination in logistics. In this context, FTL transport is used when the goods' volume exceeds a vehicle's capacity, making direct shipment possible. LTL is applied when the volume is insufficient to fill a truck, and the goods are first sent to a consolidation center before reaching the final destination. In (Schöneberg, Koberstein, and Suhl 2013), the proposed model accounts for demand uncertainty.

(Díaz-Madroño, Mula, and Peidro 2014) present a fuzzy multi-objective integer linear programming (FMOILP) model that focuses on operational procurement transport planning in an automotive supply chain, which involves a single source and a single destination while utilising FTL. The developed model aims to minimise the number of trucks employed for procurement transport and the inventory level by considering the uncertainties related to transport capacity levels.

(Hein and Almeder 2016) investigate the benefits of integrated procurement transport and production planning for supply and production processes by comparing a sequential approach, where production is firstly determined, followed by procurement transport decisions, plus an integrated approach in which decisions are jointly made. Numerical experiments reveal that the integrated planning approach leads to substantial cost savings.

(Díaz-Madroño, Mula, and Peidro 2017) present an novel MILP model that integrates procurement, production and transport planning decisions within manufacturing systems. This model extends beyond traditional material requirement planning (MRP) systems by incorporating procurement transport decisions from multiple sources and a single destination into a unified optimization framework. It considers the capacities of both material and production resources, along with procurement transport planning decisions, by employing various shipping modes such as FTL, LTL, and milk-run.

(Chitsaz, Cordeau, and Jans 2019) address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces a novel MILP model for the ARP and proposes a three-phase decomposition metaheuristic for its solution.

(Baller et al. 2022) examine the optimization of procurement transport in an automotive factory, focusing on first-tier suppliers at an aggregate level across multiple sources (570) to a single destination, using FTL, LTL, and courier express service (CES) as transportation methods. The proposed MILP approach includes a cyclic inventory constraint, truck utilization, and storage limitations.

(Hashemi-Amiri, Ghorbani, and Ji 2023) propose an integrated approach to optimize a perishable food supply chain. The model combines supplier selection, procurement and distribution transport, production scheduling, and vehicle routing to improve supply chain efficiency and resilience. It handles demand uncertainties through a distributionally robust chance-constrained model. The model is reformulated as MILP by utilizing duality and linearization techniques.

2.2 Procurement Transport with RTIs

Some models further integrate the procurement and delivery of RTIs into the procurement transport literature. In this sense, RTIs are procured to the supplier, who loads the items and sends them to customer. We extend the SC definitions previously defined to contemplate the RTIs management strategies.

model a food closed-loop supply chain network design problem incorporating returnable transport items (RTIs), product perishability, facility location and demand uncertainty. A two-stage non-linear chance-constrained programming model minimizes fixed facility, production, transport, inventory holding and RTI

costs while ensuring a target service level. It’s linearized via sample average approximation and distribution-free methods to capture forward RTI-based shipments and reverse flows for reuse. To solve large instances, an improved Lagrangian relaxation algorithm decomposes facility-location and flow decisions, outperforming CPLEX on solution quality and speed.

(Soysal et al. 2023) develop a tactical-level, multi-period closed-loop Vendor-Managed Inventory (VMI) system for RTIs in an agro-food SC. The SM SC integrates forward flows of full RTIs from a single vendor to multiple customers and reverse flows of empties back to the vendor for cleaning . All routing and inventory decisions are made centrally by the vendor under full information sharing. Sustainable plans in terms of amounts of energy spent and empty crates collected are devised and evaluated.

(Guo et al. 2024) model two-tier Container Management Centre (CMC) network . Empty RTIs are dispatched from central CMC hubs to upstream suppliers, loaded with products, carried by truck to downstream customers, then returned empty to regional CMCs for cleaning and storage before being shipped back to central hubs. A neural network is used for demand forecasting. A MILP is proposed that maximises total profit by optimally determining transport costs and the penalties applied to late RTIs returns.

(Y. Zhang et al. 2024) formulate a perishable-food inventory-routing MILP with multiple RTI types in a SM supply chain, integrating procurement of products and pickup of both filled and empty RTIs. Decisions are made on a operational level, optimally determining per period the quantities to pack, deliver, pick up, purchase and lose, as well as the vehicle routes in order to maximise the total profit of the SC that is equal to the total quality-dependent revenue minus the total costs involved, including inventory holding costs, product cost, transportation costs, and RTI purchasing and loss costs. The model is solved exactly via CPLEX solver and approximately via a tailored Kernel Search matheuristic.

2.3 Container loading

Container loading problems encompass a subset of geometric assignment problems known as cutting and packing (*C&P*) problems. They revolve around the allocation of n -dimensional orthotope items in larger n -dimensional orthotope containers. Here the aim is to optimise an objective function while taking into account two key geometric constraints: complete containment of items inside containers and lack of item overlap. A loading pattern serves as a formal representation of a solution to such an assignment problem. Dyckhoff introduced in (Dyckhoff 1990) a typology that unifies *C&P* problems using four different criteria: dimensionality of space, the objective of assignment, containers types and box type. Consequently, (Wäscher, Haußner, and Schumann 2007) develop a new typology that has become the standard in the *C&P* literature. We briefly describe the typology used in (Wäscher, Haußner, and Schumann 2007) based on five different criteria: (i) dimensionality of space; (ii) kind of assignment (input minimisation of containers for all small items and loading patterns; output maximisation to pack a subset of small items with limited container availability); (iii) assortment of small item by distinguishing if they are either a weakly (or strongly) heterogeneous assortment or identical small items ; (iv) assortment of large objects (if it is either several large objects or one large object); (v) shape of small items (circular or of another shape). Table 1 links the standard *C&P* nomenclature and the criteria of the typology according to (Wäscher, Haußner, and Schumann 2007).

The first mathematical programming model in a scientific journal to describe the container-loading problem was developed in (C. S. Chen, Lee, and Shen 1995), where the authors introduced an MILP model to address the 3D input minimisation problem by considering rotations and containers of varying dimensions. However, it does not specify the use of weakly or strongly heterogeneous boxes or containers. Additionally, real-world constraints are modelled like, for example, weight distribution.

(Junqueira, Morabito, and Sato Yamashita 2012b) propose an alternative MILP formulation to (C. S. Chen, Lee, and Shen 1995) using binary encoding of box position, which utilises normal patterns to reduce possible box coordinates by considering multiples of boxes dimensions and ensuring optimal solutions. Additionally, loading and cargo constraints are developed to ensure the static and dynamic stability of the cargo. Boxes must be supported by other box top faces or the container floor. The same authors integrate further loading restrictions in (Junqueira, Morabito, and Sato Yamashita 2012a), where they integrate multidrop constraints, which address the challenge of packing a truck with boxes for different customers at various destinations. The goal is to plan the container’s loading to minimise the time spent unloading and reloading boxes at each destination.

(W. Zhu and Lim 2012) devise a tree search-based method to solve SLOPP and SKP with 3D constraints. Their approach involves evaluating free spaces by employing a look-ahead tree search and selecting optimal placements based on the best free space-block pairs.

(Gonçalves and Resende 2013) develop a biased random-key genetic algorithm (BRKGA) designed to solve SBSBPP. The algorithm addresses geometric constraints and introduces a maximal-space representa-

tion to handle available free spaces in bins. A novel placement procedure is hybridised with the BRKGA, which optimises the order of box packing and the parameters utilised by the placement procedure to offer an effective solution approach to an NP hard problem.

(Bortfeldt and Wäscher 2013) identify constraints for $C\&P$, including item-related constraints (orientation constraints, load bearing, loading priorities), container-related constraints (weight distribution and weight limit), and cargo-related constraints (complete-shipment constraints) and load-related constraints (complexity constraints or stability constraints). The study also highlights the achievements and deficits in container loading research that pertain to these constraints.

(Paquay, Schyns, and Limbourg 2016) propose an MILP model for the MBSBPP. One noteworthy constraint is the assumption that containers are truncated parallelepiped instead of rectangular boxes. They also consider constraints, such as box rotation, container load bearing, weight distribution, stability and fragility of items.

(Mahvash, Awasthi, and Chauhan 2018) introduce a novel resolution procedure for the SBSBPP through the column generation technique, where the columns that are generated are the loading configurations of a single bin. For the pricing develop problem they develop a heuristic to find new columns with a negative reduced cost. This approach not only produces better results than the previous articles on the literature, but also gives new lower bounds.

(Silva, Toffolo, and Wauters 2019) study SSLOP models with 3D constraints. These authors compare the different approaches that exist in the literature for model packing problems based on two fundamental approaches: discretised formulation (Junqueira, Oliveira, et al. 2013) and continuous formulation (C. S. Chen, Lee, and Shen 1995) with subsequent modifications. They also indicate that discretised formulation (Junqueira, Morabito, and Sato Yamashita 2012a) performs the best for instances with similar sized boxes. However, it encounters memory limitations for large instances. For other sets of instances, the adapted formulations of (C. S. Chen, Lee, and Shen 1995) and (Paquay, Schyns, and Limbourg 2016) outperform other adaptations. Further research topics suggest including the design of new formulations by considering the limited approaches in the literature.

(Kurpel et al. 2020) introduce techniques and exact approaches to solve input minimisation and output maximisation problems for container-loading problems. Their methods consider practical constraints, such as orientation, separation of boxes and vertical stability. A set partitioning method is also employed in order to obtain upper bounds in a container-loading input minimisation problems. The authors adapt and evaluate four discretisation methods found in the existing literature and incorporate mathematical formulations to address the aforementioned constraints.

(Alonso, Alvarez-Valdes, and Parreño 2020) develop an algorithm by the greedy randomised adaptive search procedure (GRASP) to tackle a practical container-loading problem with time-sensitive constraints. Their approach addresses dynamic stability constraints while stacking products on pallets and loading them onto trucks during different time periods. Their approach provides high-quality solutions and surpasses an integer programming formulation in efficiency and effectiveness terms.

(Ali et al. 2022) conduct a comprehensive literature review by identifying constraints and solution methods for offline problems with full knowledge of the items to be packed and those tackling the online problems, where items are packed immediately when they arrive, and no prior knowledge of an item is known. The study also identifies and classifies solution approaches for offline packing into six categories: exact methods, constructive heuristics; local search and metaheuristic methods; approximation algorithms; machine learning and hyper-heuristic methods; tree search methods. (Harrath 2022) proposed a three stage heuristic based on layer generation to find good quality solutions the SBSPP problem for large instance up to 993 bins and 200000 items taking into account load balancing constraints.

(Tresca et al. 2022) develop a matheuristic to solve the SBSBPP that includes items with different IDs and different categories, to be delivered to different clients. The algorithm creates monoitem, monocategory, and mixed layers which then are used to compute the final loading configurations of the bins.

(Li, M. Chen, and Huo 2022) develop a hybrid adaptive large neighborhood search algorithm to solve the multiple container loading problem, which is to pack a given set of items with the minimum number of containers while maximizing the space utilization of them. Therefore, it is both a MBSBPP and a MSSCSP. Their approach takes into account some practical applications such as suspension constraints, rotation and weight limit.

(M. Chen, Huo, and Duan 2023) present a new approach to the 3D-BPP, introducing the 3D-BSBPP which focuses on optimizing both the design of bin sizes and the packing process. A novel Hybrid Biogeography-Based Optimization (HBBO) algorithm is proposed, integrating differential evolution and a local search strategy for enhanced performance.

One noteworthy trend in $C\&P$ is integrating loading problems with routing. This is known as capacitated vehicle-routing problem with 3D or 2D loading constraints (3L-CVRP or 2L-CVRP). (Junqueira, Morabito,

and Sato Yamashita 2012b) present a MILP model with binary encodings for 3L-CVRP by including loading, cargo and multidrop constraints. The combination of these two NP-hard problems requires using heuristics and metaheuristics for real-world instances, as seen in (Dominguez et al. 2016), (Alonso, Martinez-Sykora, et al. 2022) and (Wei et al. 2015). Another noteworthy trend is to use reinforcement learning techniques to solve $C\&P$, as in (Que, Yang, and D. Zhang 2023) and (H. Zhao, C. Zhu, and Huang 2022).

2.4 Research Gaps & Methodology

Here, we discuss the main differences between the 3D-SCPTOP model proposal and those reviewed by specifically focusing on procurement transport. The key distinction of the 3D-SCPTOP model compared to the existing literature is the incorporation of 3D constraints and vertical stability constraints to model transport capacity with a view to providing more realistic solutions and reducing the likelihood of infeasible outcomes. Table 1 presents a comparison of inventory modelling, transport modelling, and supply chain configuration between the 3D-SCPTOP model proposal and those currently in the literature. Hence, the 3D-SCPTOP approach is based on a SS supply chain configuration, and takes into account the FTL transportation mode.

Article	SC Configuration	Decision level	Transportation Mode	Transportation Capacity Constraint	Transportation Cost	Inventory Capacity Restriction	RTIs	Inventory Costs	Safety Stocks	Modelling Framework
(Guo et al. 2024)	MM	TO	FTL	UH	X	X	S—NF	X		MINLP
(Berman and Wang 2006)	MM	T	FTL—LTL	V	X			X		MINLP
(Sancak and Salman 2011)		O	LTL—FTL	V	X			X	X	MILP
(Schöneberg, Koberstein, and Suhl 2010)	MS	T	FTL—LTL	V	X	X		X		MILP
(Schöneberg, Koberstein, and Suhl 2013)	MS	O	FTL—LTL	V	X	X		X		MILP
(Díaz-Madroño, Mula, and Peidro 2014)	SS	O	FTL	UH	X			X	X	FMOILP
(Hein and Almeder 2016)	MS	O	MR	V	X			X		MILP
(Díaz-Madroño, Mula, and Peidro 2017)	MS	T	FTL—LTL—MR	V	X	X		X	X	MILP
(Meyer and Amberg 2018)	MS	T—O	MR—FTL	V	X	X		X		MILP
(Chitsaz, Cordeau, and Jans 2019)	MS	O—T	MR	V	X	X		X		MILP
(Baller et al. 2022)	MS	O	MR—FTL—LTL—CES	V—W	X	X			X	MILP
(Hashemi-Amiri, Ghorbani, and Ji 2023)	MS	T	MR	V	X					CCP
3D-SCPTOP	SS	O	FTL	3D-Constraints	X	X		X	X	MILP
(Y. Zhang et al. 2024)	MS	TO	MR	UH	X	X	M—NF	X		ILP

Abbreviations. **T** (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight), **UH** Unit Holding; **CCP** chance-constrained programming

Table 1: Comparison of procurement transport models

On the other hand, the 3D-SCTOP model diverges from those proposed in the packaging literature in several aspects. Firstly, the loading problem takes into account different time periods. Therefore, the specific container (small items or boxes in the $C\&P$ literature) that we utilise are yet to be defined, as the choice of container depends on the inventory levels of each product. Also, the objective function depends on inventory and transport costs, which are inherently connected to the loading problem at hand. Therefore, we simultaneously solve an input minimisation and output maximisation problem. Both of these are a result of transport minimisation and inventory costs. Input minimisation comes because we want to find the minimum number of trucks needed. Meanwhile, output maximization comes from the FTL and the transport costs.

Based on the literature review and the identified research gaps, our proposal is structured according to the steps depicted in Figure 1.

Article	Problem type	Resolution procedure	Orientation	Weight	Load Balance	Stability	Time-Dependent
(W. Zhu and Lim 2012)	SLOP—SKP	TSM	X	X			
(Gonçalves and Resende 2013)	SBSBPP	BRKGA	X	X		X	
(Paquay, Schyns, and Limbourg 2016)	SBSBPP	EM	X	X	X	X	
(Mahvash, Awasthi, and Chauhan 2018)	SBSBPP	CG	X	X			
(Alonso, Alvarez-Valdes, and Parreño 2020)	SSSCSP	GRASP	X	X	X	X	X
(Tresca et al. 2022)	SBSBPP	MH	X	X	X	X	
(Harrath 2022)	SBSBPP	H	X	X	X		
(Li, M. Chen, and Huo 2022)	SBSBPP—MSSCSP	HALNs	X	X		X	
(M. Chen, Huo, and Duan 2023)	BSDPP	HBBO	X	X			
3D-SCPTOP	SSSCSP & MILOPP	MH	X	X	X	X	X

Abbreviations. **TSM** (Tree search method), **BRKGA** Biased Randomized Jey Genetic Algorithm; **EM** (Exact Methods), **MH** (MatHeuristic), **CG** (Column Generation), **H** (Heuristic); **HCLP** (Hybrid Adaptive Large Neighborhood Search), **HBBO** (Hybrid Biogeography-Based Optimization)

Table 2: Comparison of C&P problems

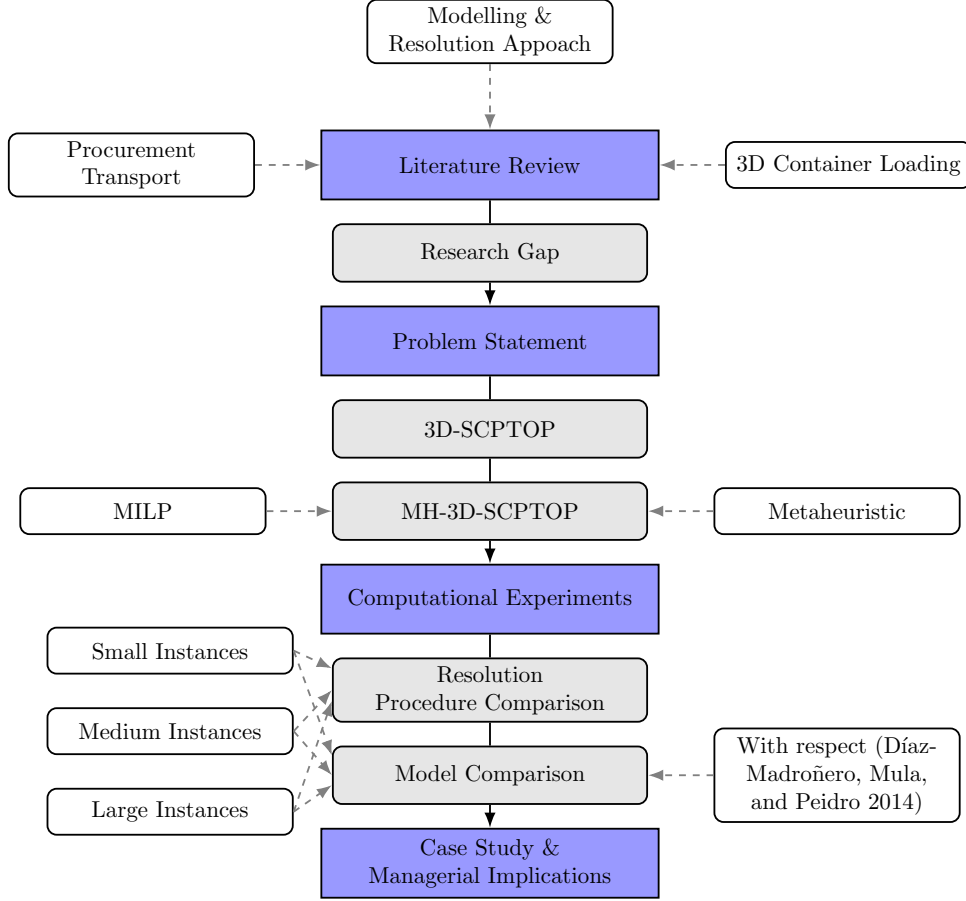


Figure 1: Research Methodology

3 Problem Description

The objective is to minimise the transportation and inventory holding costs of a manufacturing site by integrating procurement transport, RTI management and inventory decisions with full-truckload (FTL) loading constraints expressed in a stack-based form. The supply chain under study has a single source and a single destination. Products are shipped in different RTI types. RTIs are not placed individually in three dimensions but are grouped into vertical *stacks* whose 2D footprints are placed on the truck floor. Each stack type has a fixed footprint and is compatible with a subset of RTI types; each layer of a stack is homogeneous (all RTIs in a layer share a single type), while different layers of the same stack may carry different compatible RTI types. The total stack height is bounded by the truck height and the weight borne by the bottom of each stack is bounded by a crush-limit cap. Trucks carry full RTIs from the supplier to the plant and return folded empties from the plant to the supplier; folding affects RTI height, and hence the number of layers that fit in a stack, but not the stack footprint.

The assembler's demand drives the procurement process because it determines the optimal quantity of each product to procure from the supplier to the manufacturing site. Raw materials and assembled parts are produced at the supplier, packed in their corresponding RTI, and transported under FTL. Each RTI carries a single product type at any given time, and each product is shipped in a single RTI type. Once full RTIs arrive at the plant, the products are consumed and the RTIs are emptied, folded and returned to the supplier.

Stacks must not overlap and must be contained within the truck's floor; the truck's total weight is bounded; and the per-stack weight cap is respected with *period-exact* knowledge of the products loaded in each slot, so that crush-limit tests reflect the actual period's load rather than a worst-case bound. Stacks may be rotated by 90° on the truck floor. Because stacks are self-supporting by construction, no additional vertical-stability constraints are required. As FTL is the mode of transport, a dispatched truck must be full

— but a truck may be full either because it has run out of space or because it has run out of payload, so the requirement is imposed on whichever of the two resources binds. Minimum coverage stock is enforced on future demand.

Initial data.

- **Product data:** initial inventory, periodic demand (in full RTIs) at the plant and at the supplier, product-to-RTI mapping, weight of each full RTI.
- **RTI data:** height when full and when folded, weight of a folded RTI, volume.
- **Stack data:** footprint (p_h, q_h) of each stack type $h \in H$, set of compatible RTI types $R(h)$, and, for each $r \in R(h)$, number of RTIs per layer n_{hr} .
- **Transport data:** number of available trucks, truck length/width/height, truck weight capacity, per-stack weight cap, minimum truck utilisation on the binding resource.

To determine.

- A set of loading configurations (LCs), each specifying the stack-type assignment, 2D placement and orientation of its slots, and the number of layers of each compatible RTI type in each slot.
- The number of trucks used per day and the LC they employ.
- The number of full RTIs of each product loaded in each slot per period (plant-bound trucks) and the number of folded RTIs of each type loaded per truck (supplier-bound trucks).
- The resulting inventory of full and folded RTIs at both nodes.

Subject to.

- 2D non-overlap and containment of stacks on the truck floor, with allowed 90° rotation.
- Per-stack height budget.
- Per-stack crush-limit weight cap, period-exact on plant-bound trucks and deterministic on supplier-bound trucks.
- Per-truck weight capacity.
- Minimum truck utilisation, on volume or on payload, whichever binds.
- Minimum coverage stock.

In order to:

- Minimise total transportation cost.
- Minimise total inventory holding cost.

4 Mathematical Model

The 3D-SCPTOP formulation integrates stack-based truck-loading constraints with procurement transport and inventory decisions. It departs from a standard 3D container-loading formulation in two related ways. First, the 3D placement of individual RTIs inside the truck is replaced by a 2D placement of variable-composition *stacks* on the truck floor, with per-stack height and weight caps. Second, the per-period decision of which product fills each slot is retained, but the crush-limit cap is evaluated slot-by-slot using the period’s actual product mix. This yields a tighter per-stack weight bound than a worst-case-per-RTI-type abstraction, while eliminating vertical-stability constraints altogether, since stacks are self-supporting by construction.

Notational conventions. Throughout this section, $g \in G = \{s, p\}$ indexes nodes (supplier and plant), $n \in N$ products, $r \in R$ RTI types, $h \in H$ stack types, $i, j \in I$ stack-slot positions within a loading configuration, $l \in L$ loading configurations, $k \in K$ trucks and $t \in T$ planning periods. Variables whose direction is determined by definition omit the g subscript: V_{klnt} and $V_{wslot_{kilnt}}$ denote full-RTI loads on plant-bound trucks, and V_{klrt}^e denotes folded-RTI loads on supplier-bound trucks. Table 3 collects the complete nomenclature.

Table 3: Nomenclature of 3D-SCPTOP (stack-based reformulation)

Indices and sets	
G	Supplier and plant nodes, $G = \{s, p\}$.
N	Products, $n = 1, \dots, N$.
R	RTI types, $r = 1, \dots, R$.
H	Stack types, $h = 1, \dots, H$.
I	Stack-slot indices within a loading configuration, $i = 1, \dots, I $ (see eq. 4.1).
L	Loading configurations, $l = 1, \dots, L$.
K	Trucks, $k = 1, \dots, K$.
T	Planning periods, $t = 1, \dots, T$.
$M(r)$	Products carried by RTI type r , $M(r) \subseteq N$.
$R(h)$	RTI types compatible with stack type h , $R(h) \subseteq R$.
Parameters	
cK	Cost of using a truck.
lt	Lead transport time between plant and supplier (and vice versa).
i_{gn}	Initial inventory of $n \in N$ at $g \in G$ (full RTIs).
ie_{gr}	Initial inventory of folded RTIs $r \in R$ at $g \in G$.
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$.
d_{gnt}	Production plan for $n \in N$ at $g \in G$ in $t \in T$.
q_{nt}	Initial arrivals of full RTIs of product n at p in $t \in \{1, \dots, lt\}$.
q_{rt}	Initial arrivals of folded RTIs of type r at s in $t \in \{1, \dots, lt\}$.
ss_{gnt}	Security stock of n at g in t (full RTIs).
ss_{grt}	Security stock of folded RTIs r at g in t .
w_n	Weight of a full RTI carrying product n .
w_r^f	Weight of a folded RTI of type r .
pK^w	Maximum truck weight capacity.
$W^{\max}$	Per-stack crush-limit weight cap.
pK, qK, rK	Truck length, width, and height.
Γ_K	Truck volume, $\Gamma_K = pK \cdot qK \cdot rK$.
α	Minimum truck utilisation, required on whichever of volume or payload binds ($0 < \alpha \leq 1$).
p_h, q_h	Footprint length and width of stack type h .
r_{gr}	Height of one RTI of type r at node g (full at p , folded at s).
γ_{gr}	Volume of one RTI of type r at node g .
n_{hr}	RTIs of type $r \in R(h)$ per layer of stack type h .
M	A sufficiently large constant.
Decision variables – LC level	
Z_{gl}	$\in \{0, 1\}$. Equals 1 iff LC l is active in direction g .
θ_{gl}	$\in \{0, 1\}$. Selects which resource the utilisation floor is applied to: volume if 1, payload if 0.
X_{gihl}	$\in \{0, 1\}$. Equals 1 iff slot i of LC l is of stack type h in direction g .
L_{gihlr}	$\in \mathbb{Z}_{\geq 0}$. Number of layers of RTI type $r \in R(h)$ in slot i of LC l in direction g .
LX_{gihl}	$\in \{0, 1\}$. Equals 1 iff stack h 's length is along the x -axis in slot i .
LY_{gihl}	$\in \{0, 1\}$. Equals 1 iff stack h 's length is along the y -axis in slot i .
PX_{gil}, PY_{gil}	≥ 0 . Bottom-left coordinates of slot i in LC l in direction g .
A_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is to the left of slot j in LC l .
B_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is to the right of slot j in LC l .
C_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is behind slot j in LC l .
D_{gijl}	$\in \{0, 1\}$. Equals 1 iff slot i is in front of slot j in LC l .
Decision variables – runtime level	
Y_{gkt}	$\in \{0, 1\}$. Equals 1 iff truck k is used in direction g in period t .

Y_{gklt}	$\in \{0, 1\}$. Equals 1 iff truck k uses LC l in direction g in period t .
V_{klnl}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n on plant-bound truck k using LC l in period t .
V_{kilnt}^{slot}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n in slot i of plant-bound truck k using LC l in period t .
V_{klrt}^e	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r on supplier-bound truck k using LC l in period t .
I_{gnt}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of n at g in t (full RTIs).
Ie_{gtr}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of folded RTIs r at g in t .
Q_{nt}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of product n in transit to the plant in t .
Q_{rt}	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r in transit to the supplier in t .

Objective

$$\text{Min } z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} cK \cdot Y_{gkt} + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} ic_{gn} \cdot I_{gnt} \quad (1)$$

Inventory and flow balance

The inventory balance block is structurally identical to the original formulation. For completeness it is reproduced below.

$$I_{pnt} = I_{p,n,t-lt} - d_{pnt} + Q_{n,t-lt} \quad \forall n \in N, t \geq lt + 1 \quad (2)$$

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + q_{nt} \quad \forall n \in N, t \in \{2, \dots, lt\} \quad (3)$$

$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \quad \forall n \in N, t = 1 \quad (4)$$

$$Ie_{ptr} = Ie_{p,t-1,r} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \quad \forall r \in R, t \geq 2 \quad (5)$$

$$Ie_{ptr} = ie_{pr} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \quad \forall r \in R, t = 1 \quad (6)$$

$$I_{snt} = I_{s,n,t-1} + d_{snt} - Q_{nt} \quad \forall n \in N, t \geq 2 \quad (7)$$

$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \quad \forall n \in N, t = 1 \quad (8)$$

$$Ie_{str} = Ie_{s,t-lt,r} + Q_{r,t-lt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t \geq lt + 1 \quad (9)$$

$$Ie_{str} = Ie_{s,t-1,r} + q_{rt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t \in \{2, \dots, lt\} \quad (10)$$

$$Ie_{str} = ie_{sr} + q_{rt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t = 1 \quad (11)$$

$$I_{gnt} \geq ss_{gnt}, \quad Ie_{gtr} \geq ss_{gtr} \quad \forall g \in G, n \in N, r \in R, t \in T \quad (12)$$

Stack packing constraints

Exactly one stack type can be assigned to a slot, and only if its LC is active:

$$\sum_{h \in H} X_{ghl} \leq Z_{gl} \quad \forall g \in G, i \in I, l \in L. \quad (13)$$

The stack occupying slot i is oriented either with its length along the x -axis or along the y -axis:

$$LX_{ghl} + LY_{ghl} = X_{ghl} \quad \forall g \in G, h \in H, i \in I, l \in L. \quad (14)$$

The total stack height cannot exceed the truck height. The following constraint also forces $L_{ghlr} = 0$ whenever $X_{ghl} = 0$, since $r_{gr} > 0$:

$$\sum_{r \in R(h)} r_{gr} L_{ghlr} \leq rK \cdot X_{ghl} \quad \forall g \in G, h \in H, i \in I, l \in L. \quad (15)$$

Slots must not overlap. For each pair $i \neq j$ in LC l :

$$PX_{gil} + \sum_{h \in H} (p_h LX_{gihl} + q_h LY_{gihl}) \leq PX_{gjl} + M(1 - A_{gijl}) \quad (16)$$

$$PX_{gjl} + \sum_{h \in H} (p_h LX_{gjh l} + q_h LY_{gjh l}) \leq PX_{gil} + M(1 - B_{gijl}) \quad (17)$$

$$PY_{gil} + \sum_{h \in H} (q_h LX_{gihl} + p_h LY_{gihl}) \leq PY_{gjl} + M(1 - C_{gijl}) \quad (18)$$

$$PY_{gjl} + \sum_{h \in H} (q_h LX_{gjh l} + p_h LY_{gjh l}) \leq PY_{gil} + M(1 - D_{gijl}) \quad (19)$$

with the cover constraint ensuring that whenever both slots are used at least one relative-position variable is active:

$$A_{gijl} + B_{gijl} + C_{gijl} + D_{gijl} \geq \sum_h X_{gihl} + \sum_h X_{gjh l} - 1 \quad \forall g, l, i \neq j. \quad (20)$$

Containment in the truck floor:

$$PX_{gil} + \sum_{h \in H} (p_h LX_{gihl} + q_h LY_{gihl}) \leq pK + M(1 - \sum_h X_{gihl}) \quad \forall g, i, l, \quad (21)$$

$$PY_{gil} + \sum_{h \in H} (q_h LX_{gihl} + p_h LY_{gihl}) \leq qK + M(1 - \sum_h X_{gihl}) \quad \forall g, i, l. \quad (22)$$

Minimum truck utilisation. Full-truckload transport requires a dispatched truck to be full, but a truck can be full in two different ways: it may run out of space or it may run out of payload. Imposing the requirement on volume alone is not a conservative simplification, it is an infeasible one — filling a fraction α of a 97.9 m³ trailer with RTIs at 590 kg/m³ weighs 27.4 t against a 24 t cap, so no truck is dispatchable at all and the instance is empty. Summing the two ratios instead is the opposite error: it accepts a truck that is 35% full by volume and 20% by weight, which is full by neither. We therefore require a truck to reach a fraction α of *either* resource, using a binary θ_{gl} to select which one binds:

$$\sum_{i \in I} \sum_{h \in H} \sum_{r \in R(h)} \gamma_{gr} n_{hr} L_{gihl r} \geq \alpha \Gamma_K (Z_{gl} - 1 + \theta_{gl}) \quad (23)$$

$$\sum_{i \in I} \sum_{h \in H} \sum_{r \in R(h)} \omega_{gr} n_{hr} L_{gihl r} \geq \alpha pK^w (Z_{gl} - \theta_{gl}) \quad \forall g \in G, l \in L, \quad (24)$$

with $\omega_{gr} = w_r^f$ in the supplier direction and $\omega_{pr} = \min_{n \in M(r)} w_n$ in the plant direction, the lightest product being the one that makes the requirement hardest to meet. The same disjunction is imposed on each dispatched truck at runtime, on the loads V_{klnt} and V_{klrt}^e rather than on the configuration's designed capacity, since a truck may be loaded to less than the configuration allows.

Truck–LC linking and runtime loading

A truck uses at most one LC in a period, and can only use an active LC:

$$\sum_{l \in L} Y_{gklt} = Y_{gkt} \quad \forall g, k, t \quad (25)$$

$$Y_{gklt} \leq Z_{gl} \quad \forall g, k, l, t \quad (26)$$

For plant-bound trucks, full RTIs of type r loaded in slot i cannot exceed the layer capacity of the stack placed in that slot; products sharing the same RTI type partition that slot-capacity:

$$\sum_{n \in M(r)} V_{kilnt}^{\text{slot}} \leq \sum_{h: r \in R(h)} n_{hr} L_{p,i,h,l,r} + M(1 - Y_{pklt}) \quad \forall k, i, l, r, t. \quad (27)$$

Truck-level aggregation of per-slot loads:

$$V_{klnt} = \sum_{i \in I} V_{kilnt}^{\text{slot}} \quad \forall k, l, n, t. \quad (28)$$

For supplier-bound trucks, folded RTIs of type r are capped by the aggregate layer capacity of the LC in use (no slot-level tracking is needed because folded-RTI weight is deterministic in r):

$$V_{klrt}^e \leq \sum_{i \in I} \sum_{h: r \in R(h)} n_{hr} L_{s,i,h,l,r} + M(1 - Y_{sklt}) \quad \forall k, l, r, t. \quad (29)$$

Weight constraints

Per-truck weight, plant-bound:

$$\sum_{l \in L} \sum_{n \in N} w_n V_{klnt} \leq pK^w Y_{pkt} \quad \forall k, t. \quad (30)$$

Per-truck weight, supplier-bound:

$$\sum_{l \in L} \sum_{r \in R} w_r^f V_{klrt}^e \leq pK^w Y_{skt} \quad \forall k, t. \quad (31)$$

Per-stack crush-limit, period-exact on plant-bound trucks (the cap reflects the actual product mix loaded in each slot):

$$\sum_{n \in N} w_n V_{kilnt}^{\text{slot}} \leq W^{\max} \quad \forall k, i, l, t. \quad (32)$$

Per-stack crush-limit on supplier-bound trucks (deterministic in the LC since folded-RTI weight depends only on r):

$$\sum_{h \in H} \sum_{r \in R(h)} w_r^f n_{hr} L_{s,i,h,l,r} \leq W^{\max} \quad \forall i, l. \quad (33)$$

Flow aggregation

$$Q_{nt} = \sum_{k \in K} \sum_{l \in L} V_{klnt} \quad \forall n, t \quad (34)$$

$$Q_{rt} = \sum_{k \in K} \sum_{l \in L} V_{klrt}^e \quad \forall r, t \quad (35)$$

4.1 Symmetry breaking and slot-set sizing

The slots $i \in I$ of a given LC are interchangeable, so any solution admits $|I|!$ permutations of the slot index. We break this symmetry by enforcing that slot $i + 1$ is used only if slot i is used:

$$\sum_{h \in H} X_{g,i+1,h,l} \leq \sum_{h \in H} X_{g,i,h,l} \quad \forall g \in G, l \in L, i \in \{1, \dots, |I| - 1\}. \quad (36)$$

The cardinality of I is a modelling parameter. An upper bound that eliminates provably unused slot variables without cutting any feasible layout is

$$|I| \leq \left\lfloor \frac{pK \cdot qK}{\min_{h \in H} p_h \cdot q_h} \right\rfloor, \quad (37)$$

corresponding to tiling the truck floor with the smallest-footprint stack type; it serves as a tight preprocessing limit on $|I|$.

Domains

$$\begin{aligned} &Z_{gl}, \theta_{gt}, X_{gihl}, LX_{gihl}, LY_{gihl}, A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, Y_{gkt}, Y_{gkl} \in \{0, 1\}, \\ &L_{gihl}, V_{klnt}, V_{kilnt}^{\text{slot}}, V_{klrt}^e, I_{gnt}, I_{egtr}, Q_{nt}, Q_{rt} \in \mathbb{Z}_{\geq 0}, \\ &PX_{gil}, PY_{gil} \geq 0. \end{aligned}$$

5 Branch and Price Solution Approach

There are three structural difficulties with the 3D-SCPTOP model of Section 4. The first is that a priori we do not know how many loading configurations are needed, so $|L|$ must be overestimated; every configuration that turns out to be unnecessary still carries its full block of placement variables. The second is that the model replicates a complete two-dimensional packing problem $|G| \cdot |L|$ times, and each replica contributes $4 \binom{|I|}{2}$ big- M disjunctions (16)–(19) whose linear relaxation is weak. The third is that the configurations are mutually interchangeable, so the search tree is massively symmetric.

It would overstate the case to call the resulting model intractable in general, and Section 6 shows why: with a good incumbent supplied, instances spanning five periods are solved to proven optimality in seconds, even at sixty thousand variables. What defeats it is the planning horizon. The placement block is replicated per configuration rather than per period, but the truck-assignment and inventory blocks grow with $|T|$, and it is their interaction with the symmetric configuration design that explodes: at ten periods the model returns no feasible solution within half an hour. Operational procurement planning lives at horizons of weeks, so that is the regime the method has to serve, and it is the regime in which a decomposition earns its place. A column generation scheme based on branch and price is therefore developed.

The key components of the branch and price algorithm are the master problem (MP), the restricted master problem (RMP), the pricing problem (PP) and the branching policy. The MP is a reformulation of the model that exploits its decomposable structure. Inspecting Section 4, every geometric constraint (13)–(23) carries a single configuration index l , and every loading constraint (27)–(33) carries a single truck index k and a single period t ; the transport and inventory block sees them *only* through the aggregate flows Q_{nt} and Q_{rt} of (34)–(35). The model is therefore block-angular, with one independent block per dispatched truck. This suggests taking as the unit of decision neither the individual RTI nor the loading configuration, but the *complete loaded truck*.

The MP is typically too large to be solved directly because it involves a huge number of columns representing all possible truck loads. To manage this complexity we consider only a smaller subset of columns, which leads to the RMP. Solving the RMP provides an optimum for the included subset, but not necessarily for the entire column set. RMP integrality constraints are therefore relaxed to obtain dual variables, and the PP uses those duals to identify columns of negative reduced cost that should be added to the RMP. If no such column exists, the integrality-relaxed MP has been solved to optimality. If all the variables associated with the solution are integers, an integer solution to the current branch has been found. If not, the branch must be divided by adding further constraints. Figure 2 exemplifies the algorithm.

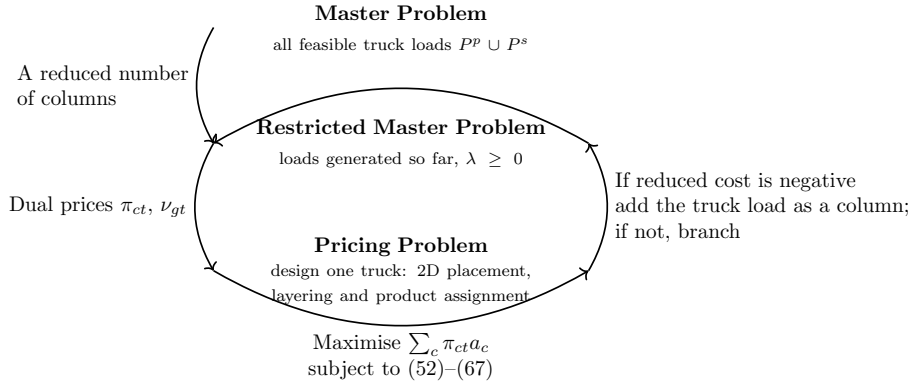


Figure 2: Branch and price scheme for 3D-SCPTOP. Every geometric, height, crush-limit, payload and full-truckload constraint is enforced inside the pricing problem, so each column is a physically loadable truck.

5.1 Formulation of the Master Problem and the Restricted Master Problem

The key idea of the set-partitioning formulation for 3D-SCPTOP is to enumerate all the feasible *truck loads* with their associated costs. A plant-bound truck load is a vector $\mathbf{a} = (a_n)_{n \in N}$ of full RTIs for which there exist a slot count $m \leq |I|$, stack types $h_1, \dots, h_m \in H$, orientations, bottom-left coordinates (x_i, y_i) , layer

counts λ_{ir} and per-slot product loads v_{in} such that

$$\text{the } m \text{ footprints, rotated as chosen, do not overlap and lie inside } pK \times qK, \quad (38)$$

$$\sum_{r \in R(h_i)} r_{pr} \lambda_{ir} \leq rK \quad \forall i \leq m, \quad (39)$$

$$\sum_{n \in M(r)} v_{in} = n_{h_i r} \lambda_{ir} \quad \forall i \leq m, r \in R(h_i), \quad (40)$$

$$\sum_{n \in N} w_n v_{in} \leq W^{\max} \quad \forall i \leq m, \quad (41)$$

$$\sum_{n \in N} w_n a_n \leq pK^w, \quad \max \left\{ \frac{\sum_n \gamma_{pp(n)} a_n}{\Gamma_K}, \frac{\sum_n w_n a_n}{pK^w} \right\} \geq \alpha, \quad (42)$$

$$a_n = \sum_{i \leq m} v_{in} \quad \forall n \in N. \quad (43)$$

Let P^p be the set of all such loads, and P^s the analogous set of supplier-bound loads $\mathbf{b} = (b_r)_{r \in R}$, built with folded heights r_{sr} , folded weights w_r^f and folded volumes γ_{sr} ; there no product assignment is needed, because the weight of a folded RTI depends only on its type.

Two features of this definition matter. First, a column carries the *per-slot* product assignment v_{in} , so the crush limit (41) is evaluated on the product mix actually riding in that slot. The column therefore reproduces the period-exact cap (32) of the compact model, and not the conservative surrogate $\bar{w}_r = \max_{n \in M(r)} w_n$ that a pool of capacity-only configurations is forced to use. Second, a column is a feasibility certificate: the placement $(h_i, x_i, y_i, \text{rotation})$ witnessing $\mathbf{a} \in P^p$ is retained, so any solution can be handed to a loading crew as a drawing and re-verified independently of the solver that produced it.

Constraint (42) is the same disjunction as (23)–(24), written on the load a truck actually carries rather than on the capacity its configuration provides. Section 6 shows that both regimes occur across the benchmark: at $v = 60$ the trailer fills by volume, at $v = 120$ it fills by weight.

Let $\lambda_{pt}^g \in \mathbb{Z}_{\geq 0}$ count the trucks dispatched in direction g and period t carrying load p , and let a_n^p and a_r^p be the RTIs of product n , respectively of type r , that load p contains. Every truck costs cK . The MP of our problem is

$$\min \quad cK \sum_{g \in G} \sum_{t \in T} \sum_{p \in P^g} \lambda_{pt}^g + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} i_{cgn} I_{gnt} \quad (44)$$

subject to

$$\sum_{p \in P^p} a_n^p \lambda_{pt}^p = Q_{nt} \quad \forall n \in N, t \in T \quad (45)$$

$$\sum_{p \in P^s} a_r^p \lambda_{pt}^s = Q_{rt} \quad \forall r \in R, t \in T \quad (46)$$

$$\sum_{p \in P^g} \lambda_{pt}^g \leq |K| \quad \forall g \in G, t \in T \quad (47)$$

(2)–(12) (inventory balances and coverage stock)

$$\lambda_{pt}^g \in \mathbb{Z}_{\geq 0} \quad \forall g \in G, p \in P^g, t \in T.$$

Objective (44) minimises transport plus inventory holding cost. Constraints (45)–(46) state that the flow of full and folded RTIs is exactly what the dispatched trucks carry, and (47) caps the fleet available per direction and period.

MP and 3D-SCPTOP have the same optimal value whenever $|L|$ is not restricted a priori. Given a 3D-SCPTOP solution, each used truck (k, t) induces a load vector satisfying (38)–(43), obtained by reading the slot data off its configuration and setting $v_{in} = V_{kilt}^{\text{slot}}$; counting trucks by load vector gives a MP solution of equal cost. Conversely, introducing one configuration per distinct load used by a MP solution and assigning trucks accordingly reconstructs a 3D-SCPTOP solution of the same cost. This equivalence is what distinguishes the present approach from a pool-based method such as PLC-SCPTOP: a predefined pool is a *restriction* and can only yield upper bounds of unknown quality, whereas MP is the same problem written differently, so any bound derived from it is a bound on the original.

The RMP replaces P^g by a subset $\bar{P}^g$ and relaxes integrality,

$$\min \quad cK \sum_{g \in G} \sum_{t \in T} \sum_{p \in \bar{P}^g} \lambda_{pt}^g + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} i_{cgn} I_{gnt} \quad (48)$$

subject to (45)–(47) restricted to $\bar{P}^g$, with $\lambda_{pt}^g \geq 0$. Dual values are obtained by solving the RMP. Let π_{nt} and π_{rt} be the duals of (45)–(46) and $\nu_{gt} \leq 0$ those of (47). The reduced cost associated with a plant-bound

load $p \in P^p$ dispatched in period t is

$$\bar{c}(p, t) = cK - \sum_{n \in N} \pi_{nt} a_n^p - \nu_{pt}, \quad (49)$$

and analogously in the supplier direction. A load with negative reduced cost can improve the current RMP solution. For each iteration we solve the subproblem of finding the load of minimum reduced cost; loads with negative reduced cost are added as new columns, the RMP is re-solved to generate new duals, and the process repeats until no further improving column is discovered.

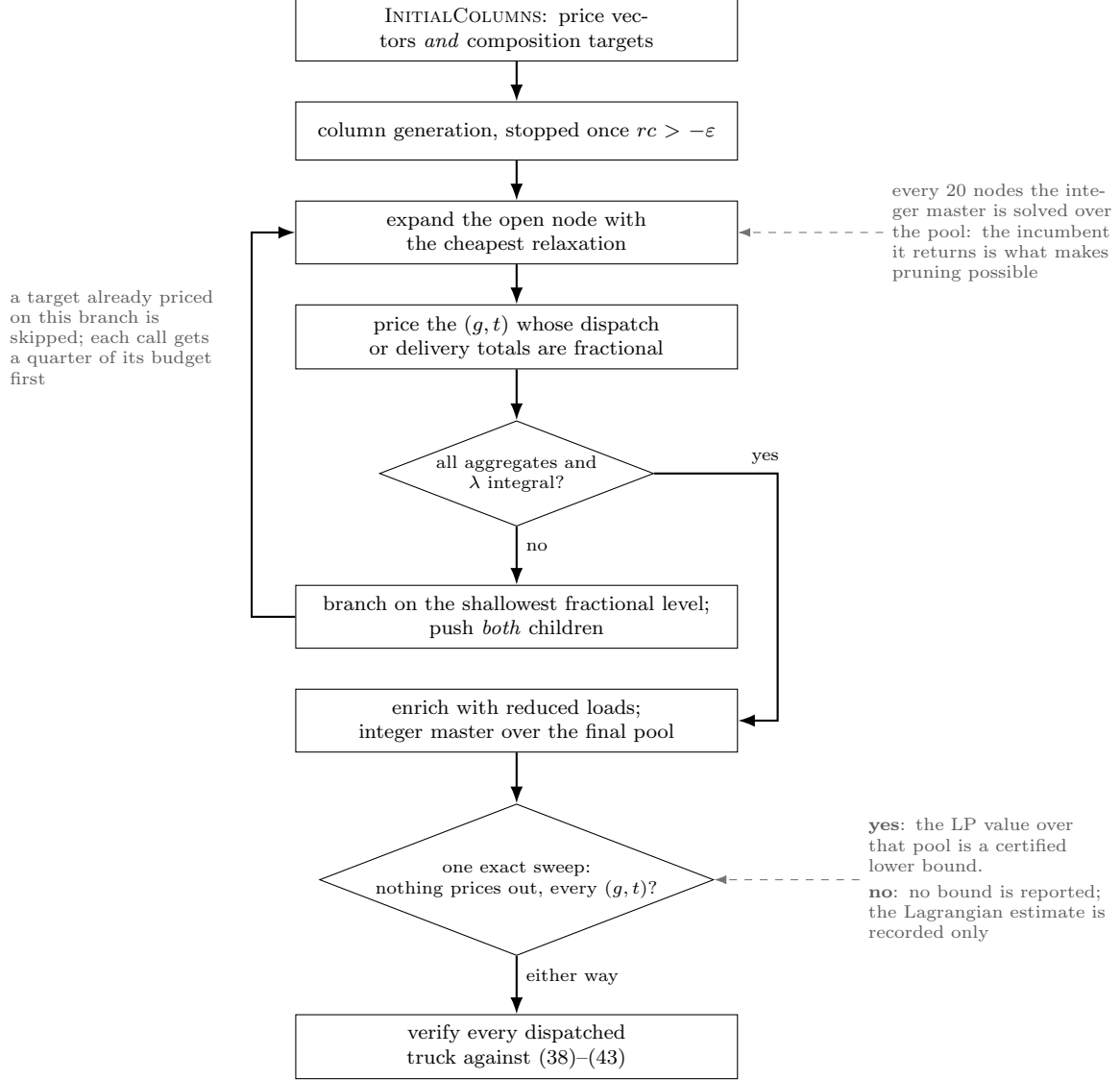


Figure 3: MATH-3D-SCPTOP end to end. Columns keep being generated *after* the branching decisions, which a price-and-branch scheme over a frozen pool cannot do. Two conventions are worth reading off the diagram: the incumbent comes from an integer master solved periodically over the pool rather than from the tree reaching integrality, and the bound is decided at a single point — either an exact sweep proves that nothing prices out anywhere, or no bound is reported at all.

5.2 Lower bounds

Care is needed here, because the natural candidate is not a bound at all. The RMP is a *restriction* of the MP: it optimises over $\bar{P}^g \subseteq P^g$, so $z_{\text{RMP}} \geq z_{\text{MP}}$. Reporting the RMP value as a lower bound is therefore valid only once column generation has converged, and wrong before then. We use two bounds accordingly.

Certified bound. When the PP has *proven* that $\zeta_{gt} := \max_{p \in P^g} \sum_c \pi_{ct} a_c^p \leq cK - \nu_{gt}$ for every g and t , no column outside $\bar{P}^g$ has negative reduced cost, so the current duals are feasible for the dual of the full MP. Weak duality then gives $z_{\text{RMP}} = z_{\text{MP}} \leq z_{3\text{D-SCPTOP}}^*$. We report such bounds as *certified*; the gap against the final integer solution is then a genuine optimality gap, and not a gap with respect to a pool.

Lagrangian bound. Certification requires the two-dimensional packing subproblem to be closed to proven optimality, which is not always possible inside a sensible time limit. Dualising (45)–(46) with the current π leaves a subproblem in which each column enters with reduced cost (49); since the number of trucks an optimal solution can dispatch in total is bounded by $\bar{X}$, the correction

$$z_{\text{MP}} \geq z_{\text{RMP}} + \bar{X} \cdot \min_{g \in G, t \in T} \min\{0, cK - \zeta_{gt} - \nu_{gt}\} \quad (50)$$

holds at every iteration, converged or not. Two choices make (50) usable rather than vacuous. The value substituted for ζ_{gt} is the solver’s own dual bound on the PP, never its incumbent, so a subproblem stopped early still yields a valid verdict. And $\bar{X}$ must be a genuine bound on the fleet actually used: the textbook form charges $|K|$ separately for every direction and period, which here is roughly an order of magnitude too pessimistic and drove the reported gaps above 80%. Because transport cost alone cannot exceed the objective, any incumbent $\bar{z}$ gives $\bar{X} = \bar{z}/cK$, which is both valid and tight enough to be informative.

What we report. We report the certified bound, and when certification fails we report no bound at all; Figure 4 states the rule exactly. The Lagrangian correction (50) is computed and recorded, but not published. In our implementation it repeatedly returned values above solutions we could exhibit: on one instance it gave 21552.23 while the pool of that same run admits a slack-free plan, verified truck by truck, costing 21502.80. What makes this dangerous rather than merely loose is that no check inside the run detects it — the run’s own restricted master lies above the true optimum too, so the estimate looks consistent from the inside, and only an externally exhibited plan exposes it. A quantity that cannot be falsified from within the method is not one we are willing to print as a bound.

Two ways a certificate can be void. A certificate is only as good as the exhaustiveness of the check behind it, and two implementation details silently broke ours. First, the sweep skipped any (g, t) whose commodity duals were all non-positive, on the argument that nothing there can pay for a truck. The reduced cost is $cK - \nu_{gt} - \sum_c \pi_{ct} a_c$: non-positive duals bound the sum by zero, so the argument establishes only $rc \geq cK - \nu_{gt}$ and holds just when $cK - \nu_{gt} \geq 0$. The test never looked at ν . A pair skipped this way contributes neither a proof nor an objection, so the certificate was claimed over pairs nobody had examined — and it is the only path by which that can happen, since a subproblem stopped on time already reports itself unproven. Second, when the pricing model is infeasible under a threshold constraint, the oracle returned a bound of zero. Infeasibility there proves only that no column beats the threshold, so the *threshold* bounds ζ_{gt} ; reporting zero understates ζ_{gt} and inflates (50). Both defects push the bound the same way, upwards, which is where every invalid bound we observed sat.

When certification is attempted. Certifying at the root is wasteful, because the tail of column generation buys almost nothing. Over ten recorded runs, stopping once the best reduced cost exceeded -0.2 gave up 0.00 of LP value in nine of them and 0.59 in 18795 in the tenth, while saving between 62 and 764 seconds — time that buys tree nodes, which is where the columns that change the fleet actually come from. We therefore stop on that tolerance and spend a single exact sweep at the end, over the final pool. If nothing prices out there and every (g, t) returned a bound, the LP value of that pool is a valid lower bound. It is weaker than an unconverged root value, since more columns can only lower the LP; unlike it, it is true.

Rounding the duals. The prices π are irrational in general and must be rounded before entering an integer-coefficient subproblem. We round them *upwards*, so the rounded objective dominates the true one and the returned bound remains an over-estimate of ζ_{gt} , as both bounds above require. Rounding to nearest — the obvious choice — makes the bound a slight under-estimate and silently invalidates the certificate; in our first implementation it returned 1000.1 against a threshold of 1000.0 on instances where the reduced cost was exactly zero, and convergence was never declared.

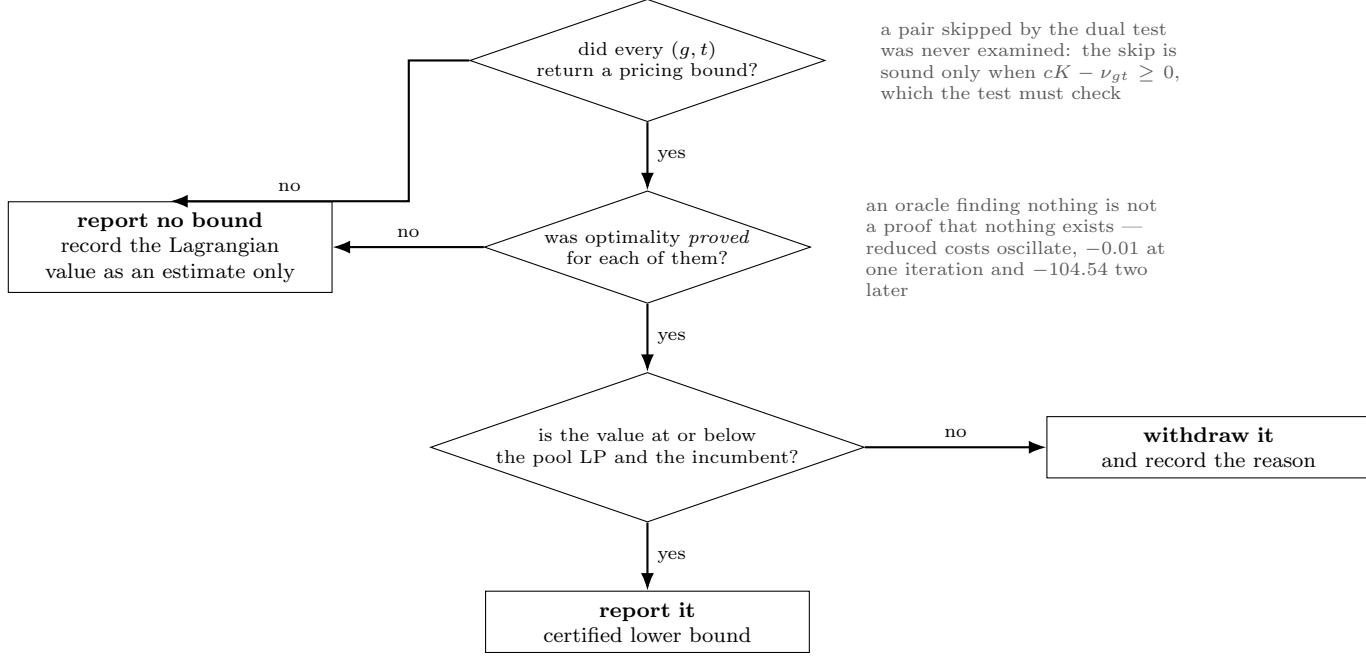


Figure 4: What may be reported as a lower bound. A quantity above a solution one can exhibit is not a loose bound but a false statement, and it inflates every optimality gap computed from it. The final test is a falsifiability check rather than a proof step: on one instance the Lagrangian value 21552.23 survived every internal consistency check while that run’s own pool admitted a verified, slack-free plan costing 21502.80 — the restricted master lies above the true optimum too, so nothing inside the run contradicts it.

5.3 Pricing Subproblem

We solve the pricing problem with a constraint programming approach. The indices and parameters are the same as in the model proposed before, but the decision variables do not depend on the configuration index. Let c range over products when $g = p$ and over RTI types when $g = s$, let $\rho(c)$ be the RTI type carrying c , and let S be the slot bound (37). Table 4 describes the new decision variables.

Table 4: Nomenclature of the pricing subproblem

Decision variables	
u_i	Binary variable indicating whether slot i is used in the new load
χ_{ih}	Binary variable indicating whether slot i holds a stack of type h
ϱ_{ih}	Binary variable indicating whether that stack is rotated by 90°
δ_i^x, δ_i^y	Footprint of slot i along the x and y axes
x_i, y_i	Bottom-left coordinates of slot i on the truck floor
ℓ_{ir}	Number of layers of RTI type r in slot i
v_{ic}	Number of RTIs of commodity c loaded in slot i
a_c	Number of RTIs of commodity c in the new load
θ	Binary variable indicating whether the full-truckload requirement is met on volume rather than on weight

The objective is to find new truck loads with negative reduced cost. Dropping the constant $cK - \nu_{gt}$ of (49), that is

$$\max z = \sum_c \pi_{ct} a_c \quad (51)$$

subject to

$$\sum_{h \in H} \chi_{ih} = u_i, \quad \varrho_{ih} \leq \chi_{ih} \quad \forall i \leq S, h \in H \quad (52)$$

$$\delta_i^x = \sum_h (p_h \chi_{ih} + (q_h - p_h) \varrho_{ih}) \quad \forall i \leq S \quad (53)$$

$$\delta_i^y = \sum_h (q_h \chi_{ih} + (p_h - q_h) \varrho_{ih}) \quad \forall i \leq S \quad (54)$$

$$\text{NoOVERLAP2D} \left(\{ [x_i, x_i + \delta_i^x] \times [y_i, y_i + \delta_i^y] \}_{i: u_i=1} \right) \quad (55)$$

$$x_i + \delta_i^x \leq pK, \quad y_i + \delta_i^y \leq qK \quad \forall i \leq S \quad (56)$$

$$\sum_{i \leq S} \sum_h p_h q_h \chi_{ih} \leq pK \cdot qK \quad (57)$$

$$\ell_{ir} \leq \lfloor rK/r_{gr} \rfloor \sum_{h: r \in R(h)} \chi_{ih} \quad \forall i \leq S, r \in R \quad (58)$$

$$\sum_{r \in R} r_{gr} \ell_{ir} \leq rK \quad \forall i \leq S \quad (59)$$

$$\sum_{c: \rho(c)=r} v_{ic} = n_{hr} \ell_{ir} \quad \forall i \leq S, r \in R \quad (60)$$

$$\sum_c w_c v_{ic} \leq W^{\max} \quad \forall i \leq S \quad (61)$$

$$a_c = \sum_{i \leq S} v_{ic} \quad \forall c \quad (62)$$

$$\sum_c w_c a_c \leq pK^w \quad (63)$$

$$\sum_c \gamma_{g\rho(c)} a_c \geq \alpha \Gamma_K \quad \text{if } \theta = 1 \quad (64)$$

$$\sum_c w_c a_c \geq \alpha pK^w \quad \text{if } \theta = 0 \quad (65)$$

$$\sum_{i \leq S} \sum_h p_h q_h \chi_{ih} \geq \alpha pK \cdot qK \quad \text{if } \theta = 1 \quad (66)$$

$$\sum_h h \chi_{ih} \geq \sum_h h \chi_{i+1,h} \quad \forall i < S. \quad (67)$$

Constraints (52)–(62) restrict the loads used in the pricing problem to feasible ones, reproducing (38)–(43). Equation (57) is a redundant area cut, and (66) its counterpart under the volume regime: a slot contributes at most (footprint area) $\times rK$ of RTI volume, so the utilisation floor implies a floor on the occupied floor area. The latter is what allows the solver to find a *first* feasible layout quickly, which for tight instances is harder than optimising once one is known. Constraint (67) breaks the symmetry among interchangeable slots by ordering them on the stack-type index.

The subproblem is stated as a constraint program because its hard core is the disjunctive non-overlap relation (55), which a CP solver propagates natively through a global constraint, whereas a MILP must encode it with the four big- M disjunctions (16)–(19) and accept the weak relaxation they entail. Section 6 reports the resulting difference.

Making the oracle affordable. Pricing dominates the running time of the tree completely: on a representative run it accounted for 1451 of 1453 seconds, against two seconds for every master LP solved at every node, and a comparable figure for the incumbent heuristic. Figure 5 shows the oracle and its two objectives. Three measures bring its cost down without changing what is solved. A node prices only the directions and periods whose dispatch or delivery totals are fractional, those being where the relaxation is interpolating between columns it already holds. A node's target is derived from the master's residual, which barely moves between a parent and its child, so a target already priced on the same branch is skipped rather than solved again. And each call receives a quarter of its budget first, the full budget only if the short call returns nothing. Together these took a node from about 28 seconds to a few milliseconds, and raised the number of nodes explored within a fixed budget by more than an order of magnitude.

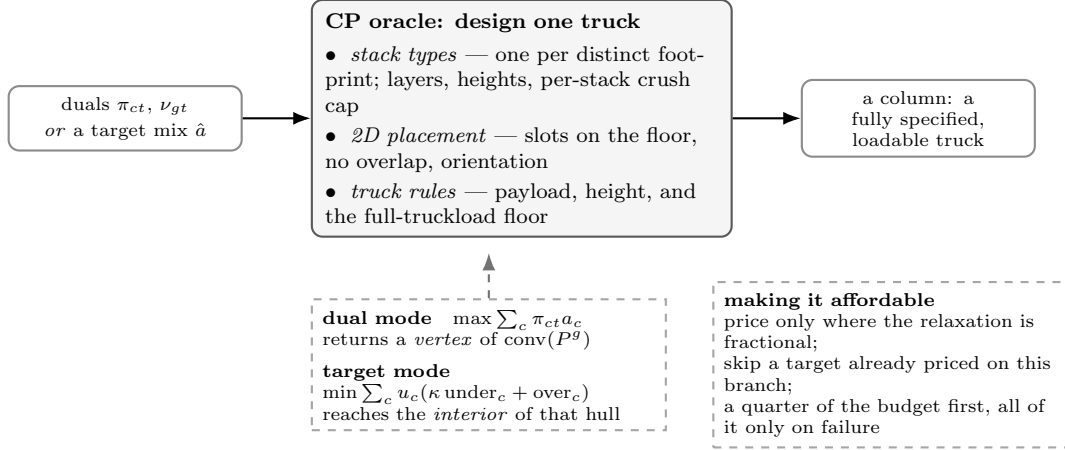


Figure 5: The pricing subproblem. Geometry, height, crush limits, payload and the full-truckload rule are enforced *inside* the oracle, so any column it returns is a physically loadable truck. In dual mode it maximises the dual-weighted load and therefore returns a vertex of $\text{conv}(P^g)$; in target mode it minimises a volume-weighted asymmetric deviation from a wanted composition, which is how loads in the interior of that hull are obtained (Figure 6). Pricing accounted for 1451 of 1453 seconds of one search, so the measures on the right are what make a node affordable.

5.4 Initial Columns

Column generation converges from any starting set, but a good one shortens it and leaves the integer master more freedom to combine loads. We initialise P^g by solving (51)–(67) under a family of demand-derived price vectors rather than duals: the unit vector, which maximises the RTI count; every single-commodity vector $\mathbf{e}_c$, which yields pure loads and keeps the master flexible; the aggregate demand mix and each per-period demand mix; and a set of random draws from the simplex for diversification.

A pool built from price vectors alone is *extreme by construction*, as Figure 6 shows on a real pool. For any weights, (51) attains its optimum at an extreme point of $\text{conv}(P^g)$, so no demand-derived vector — aggregate, per-period, random or pure — can return a load from the interior of that hull. Yet interior mixes are precisely what an integer plan needs when demand does not decompose into full trucks, and on one instance the entire supplier side of the pool consisted of two pure loads for this reason. We therefore also seed by *composition*: given a wanted mix $\hat{a}$, the oracle minimises the volume-weighted asymmetric deviation $\sum_c u_c(\kappa \text{ under}_c + \text{over}_c)$ with $\kappa > 1$, subject to the same geometry, payload, crush and full-truckload constraints. Because the target enters only the objective, a mix too small to fill a truck degrades into the closest fillable load rather than into infeasibility — which is what the last truck of a period needs. Seeding the per-period demand mixes this way took that supplier side from two pure loads to five mixed ones.

Two devices make this robust. A constructive *grid layout* — tile the floor with a single stack type in its better orientation and fill each slot to its height, weight and crush limit — is supplied to the CP solver as a hint and retained as a fallback column. For the near-tiling footprints of a real RTI catalogue it is already close to the best attainable utilisation, and it guarantees that no direction is ever left without columns. And any pricing call that returns nothing is retried with a longer limit before being believed, since an empty answer usually means the solver was starved of time rather than that no load exists.

Before the integer master is solved, the pool is enriched with *reduced* loads obtained by unloading RTIs from existing columns. Removing RTIs from a valid layout cannot break the geometry, the height budget, a crush limit or the payload, so every reduced load that still clears the full-truckload floor is itself feasible. This matters because columns priced at the LP optimum are close to full trucks, and integer combinations of full trucks are too coarse to land inside the narrow delivery band left open by the supplier’s coverage stock. The LP hides this, because it can interpolate between columns; the integer master cannot.

5.5 Branching

Branching on the master’s own variables is the obvious rule and the wrong one here. Fixing the λ_{pt}^g with the largest fractional part removes one load from the relaxation, but a Dantzig–Wolfe master of this kind holds

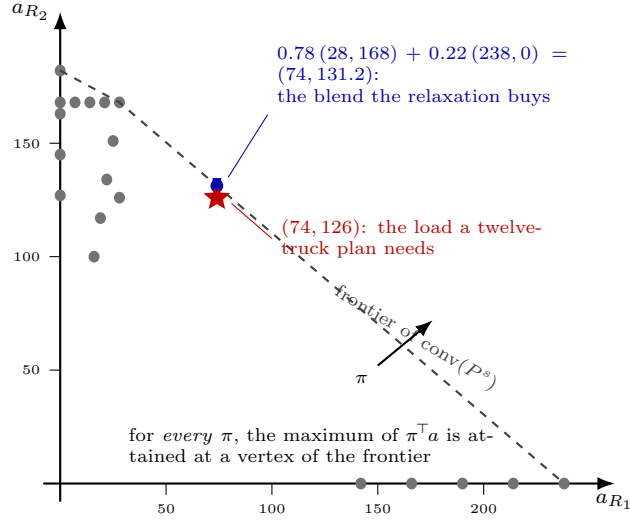


Figure 6: Why the load a cheaper plan needs is never priced. Dots are the supplier-bound loads obtained from a sweep of demand-derived price vectors; the dashed line is the upper-right frontier of their convex hull. The pricing objective is linear, so for *any* dual vector π its optimum is attained at a vertex of that frontier. The load $(74, 126)$ lies strictly below it — it is dominated componentwise by the blend $0.78(28, 168) + 0.22(238, 0) = (74, 131.2)$ — so no choice of duals returns it. The relaxation buys that blend for one truck, while an integer plan must pay for both of the trucks it mixes. Branching down withdraws a dominating vertex, which moves the duals and makes the interior reachable.

many columns that differ only in detail, so the relaxation answers by shifting the same fractional weight onto a near-identical column. Run to exhaustion on one benchmark instance with the pool frozen, that rule branched 3000 times and reached depth 1971 without ever producing an integral relaxation.

We therefore branch on aggregates first, taking the shallowest level that is fractional:

1. the *dispatch total* $\sum_p \lambda_{pt}^g$ of a direction and period. It is integer in every feasible plan and has no symmetric twin: every column of that period lies on one side of the row, so the dichotomy partitions plans rather than one variable's value.
2. the *delivery totals* Q_{nt} and Q_{rt}^* . These count physical units and are integer in any plan, although the relaxation leaves them continuous. They are original variables, so bounding one tightens a row that already carries a dual and the pricing subproblem keeps its structure.
3. the column variables λ_{pt}^g , as a last resort.

Figure 7 summarises the rule. The two aggregate levels absorb the fractionality almost entirely: on a representative run, 61 of 90 branchings were on delivery totals, 29 on dispatch totals, and none on a column variable.

Both children are kept and the node with the cheapest relaxation is expanded next, so a bad first move is recovered rather than carried to the bottom. Every node re-prices, and that is what makes the tree worth its cost. The pricing objective (51) is linear, so for *any* dual vector its optimum is an extreme point of $\text{conv}(P^g)$. A balanced load lying inside that hull is dominated by a fractional combination of extreme loads and can therefore never be priced, however the duals move — the relaxation buys the combination for one truck, while an integer plan must pay for each of them. Branching down withdraws the dominating columns, the duals move, and the oracle is asked for a load that does not yet exist. On the hardest of our instances the search derived in this way the only mixed supplier load a twelve-truck plan admits, which column generation had not produced in 978 columns; with it, a plan one truck smaller became reachable.

Reaching integrality by branching alone remains slow, so the integer master is solved over the current pool every 20 nodes and its solution kept as incumbent. It costs milliseconds at these pool sizes, and without it the open list only grows, there being no incumbent against which to prune.

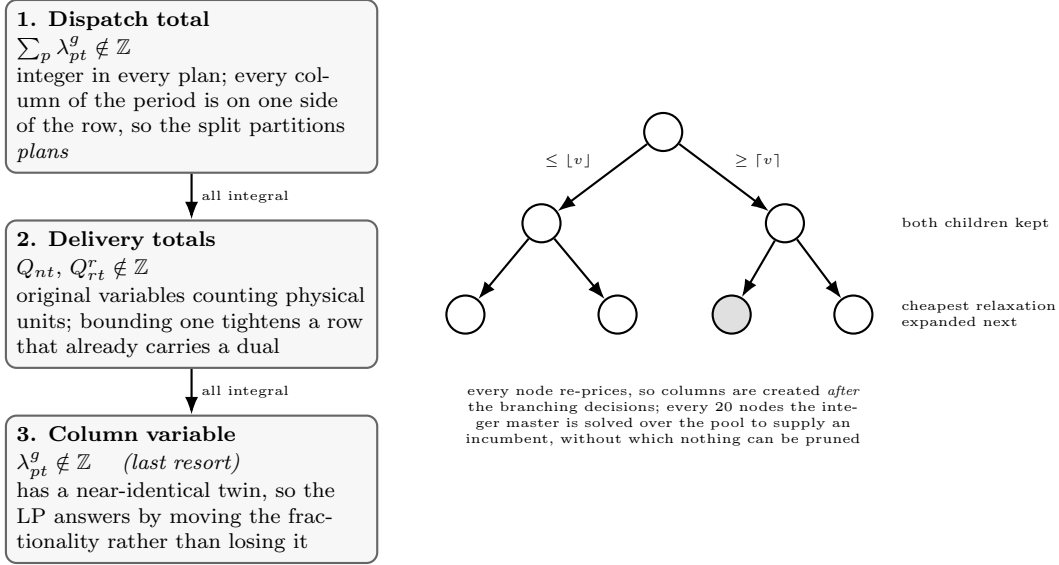


Figure 7: Branching in MATH-3D-SCPTOP. The shallowest fractional level is branched on, so the search works on aggregates that are integral in any feasible plan before it touches the master’s own variables. Under the column-variable rule alone, one instance was branched 3000 times to depth 1971 without ever reaching an integral relaxation; with the hierarchy, 90 branchings on a representative run split 61 on delivery totals, 29 on dispatch totals and none on a column variable.

5.6 Two corrections to the compact model

Implementing 3D-SCPTOP surfaced two defects worth recording, both invisible at $lt = 1$.

Inventory recursion under a lead time. Constraints (2) and (9) carry inventory forward from period $t - lt$, whereas inventory accumulates one period at a time and only the *arrival* is delayed. They must read

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + Q_{n,t-lt} \quad \forall n \in N, t \geq lt + 1, \quad (68)$$

$$I_{estr} = I_{e,s,t-1,r} + Q_{r,t-lt} - \sum_{n \in M(r)} d_{snt} \quad \forall r \in R, t \geq lt + 1. \quad (69)$$

With $lt = 1$ the two versions coincide, which is why the defect survived; with $lt \geq 2$ the original silently discards $lt - 1$ periods of stock.

Coverage stock on folded RTIs. Requiring $I_{egtr} \geq ss_{gtr}$ at *both* nodes, as (12) does, forces the plant to hoard empties it has no use for and can render an otherwise feasible instance infeasible when the RTI fleet is tight. The operationally meaningful requirement is that the *supplier* hold enough folded RTIs to cover its production, so we impose the bound at $g = s$ only.

6 Computational Experiments

The purpose of the experiments is not to establish optimality. For procurement transport planning at an operational level, what matters is whether a procedure returns, in a usable amount of time, a plan that a loading crew can actually execute and that improves on what the company does today. The experiments are organised around those two questions, and a third that turns out to dominate both: whether a model that abstracts the loading geometry away produces plans that can be executed at all.

All calculations use Python. The master and restricted master problems, the volumetric benchmark and the compact model are solved with Gurobi 12; the pricing subproblem (51)–(67) with CP-SAT from OR-Tools 9.12. Tests run on a 16-core machine with 30 GB of RAM. A relative gap of 2% is set on every integer program.

Algorithm 1 Branch and price for 3D-SCPTOP

```
1:  $\bar{P} \leftarrow \text{INITIALCOLUMNS}()$   $\triangleright$  price vectors and composition targets, Section 5.4
2:  $\bar{P} \leftarrow \text{COLUMNGENERATION}(\bar{P}, \text{stop when } rc > -\varepsilon)$   $\triangleright$  the tail buys almost nothing, Section 5.2
3:  $\mathcal{O} \leftarrow \{\text{root}\}$ ;  $\text{UB} \leftarrow \infty$ ;  $\text{LB} \leftarrow 0$ 
4: while  $\mathcal{O} \neq \emptyset$  and time remains do
5:    $n \leftarrow$  node of  $\mathcal{O}$  with the cheapest relaxation; remove it from  $\mathcal{O}$ 
6:    $\pi, \nu, \lambda \leftarrow \text{SOLVERMP}(\bar{P} \mid n)$ 
7:   if infeasible or its value  $\geq \text{UB}$  then
8:     continue  $\triangleright$  pruned
9:   end if
10:  for all  $(g, t)$  whose dispatch or delivery total is fractional do
11:     $\hat{a} \leftarrow$  residual of  $(g, t)$ ; skip if  $\hat{a}$  was already priced on this branch
12:     $p \leftarrow \text{PRICE}(\hat{a}, \tau_{\text{short}})$ , retried at  $\tau_{\text{full}}$  only if empty
13:    append  $p$  to  $\bar{P}$  if new  $\triangleright$  columns are created after the branching decisions
14:  end for
15:  if  $n$  is a multiple of the heuristic period then
16:     $\text{UB} \leftarrow \min\{\text{UB}, \text{SOLVERMP}(\bar{P}) \text{ as an integer program}\}$   $\triangleright$  supplies the incumbent that
    makes pruning possible
17:  end if
18:   $\lambda \leftarrow$  re-solve the relaxation, columns having been added
19:  if  $\lambda$  and all aggregates are integral then
20:     $\text{UB} \leftarrow \min\{\text{UB}, \text{value of } \lambda\}$ ; continue
21:  end if
22:  choose the shallowest fractional level: dispatch total, else delivery total, else  $\lambda_{pt}^g$   $\triangleright$  Section
    5.5
23:  push both children  $\leq \lfloor v \rfloor$  and  $\geq \lceil v \rceil$  onto  $\mathcal{O}$ 
24: end while
25:  $\bar{P} \leftarrow \bar{P} \cup \text{REDUCEDLOADS}(\bar{P})$ 
26:  $\text{UB} \leftarrow \min\{\text{UB}, \text{SOLVERMP}(\bar{P}) \text{ as an integer program}\}$ 
27: if one exact sweep proves  $rc \geq 0$  for every  $(g, t)$  then
28:    $\text{LB} \leftarrow$  LP value over the final  $\bar{P}$   $\triangleright$  certified; otherwise no bound is reported
29: end if
30: verify every dispatched column against (38)–(43)
31: return  $(\text{UB}, \text{LB})$ 
```

6.1 Instance generation

Thirty-six instances are artificially generated as a full factorial over the four explanatory variables of the study: the number of products $|N| \in \{4, 8\}$, the horizon $|T| \in \{5, 10\}$, the number of RTI types $|R| \in \{2, 3, 4\}$ and the volumetric granularity $v \in \{60, 90, 120\}$. Instances are named $I_{-|N|_{-}|T|_{-}|R|_{-}v}$.

- **Container dimensions:** footprints are drawn from the industrial catalogue $\{1200 \times 800, 1200 \times 1000, 1200 \times 1600, 2400 \times 1200\}$ mm. Heights are quantised to 50 mm and confined to $[400, 1500]$ mm so that at least two layers fit under the trailer roof. Whenever $|R| \geq 3$ several RTI types share a footprint, reproducing a real catalogue in which container families sit on a common pallet base; this is what allows a stack to mix types across layers.
- **Container volume:** v is the number of erected RTIs that fill one truck by volume, and every RTI type is generated with volume at least Γ_K/v . Larger v means smaller RTIs and a harder placement subproblem.
- **Demand:** generated per product and period by deviating $\pm 10\%$ from a mean calibrated so that roughly 2.5 truckloads are consumed per period. Calibrating the mean keeps difficulty comparable across the factorial instead of letting it drift with the RTI size.
- **Weights:** the content weight of a full RTI scales with its volume and varies by $\pm 25\%$ between products sharing an RTI type, which is what makes the per-stack crush limit depend on the period's product mix rather than on the RTI type alone.

The truck is a standard semi-trailer, $pK \times qK \times rK = 13600 \times 2400 \times 3000$ mm, payload $pK^w = 24$ t, per-stack crush limit $W^{\max} = 1200$ kg, minimum utilisation $\alpha = 0.55$ and cost $cK = 1000$ per dispatch. Lead time is $lt = 1$ and coverage stock one period of demand.

Sizing the RTI fleet. One generation parameter deserves comment because it is easy to get wrong and the symptom is misleading. Empties return by the truckload, and a return truck leaves only once the plant has accumulated enough folded RTIs to clear the full-truckload floor. Since folded RTIs pack very densely — two to three hundred to a trailer against an accumulation rate of a few dozen per period — that takes several periods, during which the supplier must live off its own folded stock. If the fleet is sized only for the lead time, the linear relaxation remains feasible by dispatching fractional trucks while the integer problem has no solution at all. We therefore size the supplier’s folded stock to cover $lt + SC + \lceil \alpha \Gamma_K / (\gamma_{sr} \cdot \text{empties per period}) \rceil$ periods, which is the standard closed-loop argument that an RTI fleet must cover the cycle time.

6.2 Methods compared

- **MATH-3D-SCPTOP** — Algorithm 1.
- **Company procedure (PLC-SCPTOP)** — the master problem over a fixed pool of manually derived loading configurations, one regular grid per container type, as produced today by a spreadsheet.
- **SCPTOP** — the volumetric benchmark of Appendix A, identical to the master in every respect *except* that truck capacity is volumetric. The comparison therefore isolates the effect of the loading representation.
- **Compact 3D-SCPTOP** — solved directly, on the instances where that is still possible, to measure the matheuristic’s true optimality gap without a bounding argument.

Every truck that MATH-3D-SCPTOP dispatches is re-verified from scratch against (38)–(43) by a routine that shares no code with the solver that produced it. Every truck that SCPTOP dispatches is submitted to an independent constraint programme asking whether that exact set of RTIs can be stacked and placed in a trailer at all.

6.3 Results

Table 5: **Preliminary: 5 of 36 instances complete; this table is not final.** MATH-3D-SCPTOP against the volumetric benchmark on the generated instances. *unloadable* counts the truck loads SCPTOP dispatches that a constraint programme proves cannot be placed in a trailer, out of all loads it dispatches. Δz is the cost of MATH-3D-SCPTOP relative to SCPTOP.

Instance	MATH-3D-SCPTOP				SCPTOP (volumetric)					Δz
	z	trucks	loadable	time (s)	z	trucks	unloadable	loadable	time (s)	
I.4.10.2.120	62 551	45	no	1 575	53 441	36	31 / 36	0	1	17.0%
I.4.10.2.60	44 099	34	no	2 020	31 910	22	20 / 22	0	0	38.2%
I.4.10.2.90	45 302	29	all	1 545	42 315	26	19 / 26	0	0	7.1%
I.4.10.3.120	58 459	42	all	1 745	47 916	32	24 / 32	0	0	22.0%
I.4.10.3.60	35 152	24	all	1 922	30 909	20	18 / 20	0	1	13.7%

6.4 Discussion

A volumetric capacity is not a conservative approximation. Across the 36 instances SCPTOP dispatches 723 truck loads. A constraint programme proves that 503 of them — 70% — cannot be placed in a trailer at all: there is no assignment of their containers to stacks that respects the height budget and the crush limit and whose footprints fit on the floor without overlap. Only 17 loads are demonstrably loadable; the rest are undecided within the time limit. SCPTOP’s plans are on average 17% cheaper than MATH-3D-SCPTOP’s, and that entire margin is bought by loading trucks beyond what geometry allows. A planner handed such a plan discovers the shortfall on the loading dock.

The effect is strongly modulated by container size. At $v = 60$, where each RTI occupies about a sixtieth of the trailer, 93% of the volumetric loads are impossible and the apparent saving rises to 33%; at $v = 90$ it

Table 6: Results by instance factor. *unloadable* is the share of SCPTOP truck loads proven impossible to place.

Group	trucks (MATH)	trucks (SCPTOP)	unloadable	Δz	columns	time (s)
$ N = 4$	34.8	27.2	82%	19.6%	653	1 761
$ T = 10$	34.8	27.2	82%	19.6%	653	1 761
$ R = 2$	36.0	28.0	83%	20.8%	577	1 713
$ R = 3$	33.0	26.0	81%	17.9%	766	1 833
$v = 60$	29.0	21.0	90%	26.0%	490	1 971
$v = 90$	29.0	26.0	73%	7.1%	752	1 545
$v = 120$	43.5	34.0	81%	19.5%	766	1 660
all	34.8	27.2	82%	19.6%	653	1 761

Table 7: MATH-3D-SCPTOP against the compact 3D-SCPTOP model, warm-started with the matheuristic’s plan, solved to a zero tolerance and given 30 minutes with $|L| = 3$ configurations per direction. Rows marked $\dagger$ are proven optimal; on the rest the compact model stopped on the time limit, so the last column is a gap to an incumbent rather than to an optimum.

Instance	compact 3D-SCPTOP					MATH-3D-SCPTOP		
	z	gap	vars	cons	time (s)	z	time (s)	gap to ref.
S_2_2_2	3 316	2.4%	11 908	16 302	1 800	4 244	624	27.99%
S_2_2_3	3 105	2.7%	12 262	17 330	1 800	3 105	622	0.00%
S_2_3_2	7 967	12.8%	12 672	18 130	1 800	7 966	855	-0.02%
S_2_3_3	7 001	0.3%	19 173	27 952	1 800	7 001	417	0.00%
S_3_2_2	3 170	0.2%	12 586	17 008	1 800	3 176	663	0.21%
S_3_2_3	3 502 †	0.0%	19 072	26 142	11	3 504	616	0.03%
S_3_3_2	8 166	11.5%	13 689	19 189	1 800	8 158	901	-0.10%
S_3_3_3	6 123 †	0.0%	20 442	29 263	14	6 127	424	0.06%
I_4_5_2_60	14 760	13.4%	26 490	40 991	1 800	14 730	959	-0.20%
I_4_5_3_90	21 503	0.0%	43 017	72 970	1 800	21 684	797	0.84%
I_8_5_2_90	18 816	0.1%	67 674	92 082	1 800	20 855	959	10.83%
I_4_10_2_90	<i>none</i>	100.0%	68 904	112 632	1 800	45 343	1 194	–%

Table 8: Pricing technology. The same matheuristic is run with the subproblem solved as a constraint programme (CP-SAT) and as a MILP with big- M non-overlap (Gurobi), under identical budgets.

Instance	oracle	z	trucks	columns	CG iters	time (s)
I.4_10_2_90	cpsat	45 401	29	692	1	822
	gurobi	45 472	29	265	1	1 033
I.4_10_4_120	cpsat	51 129	35	1254	2	790
	gurobi	50 966	35	480	4	710
I.4_5_2_60	cpsat	14 730	9	333	2	715
	gurobi	14 736	9	276	2	767
I.4_5_3_90	cpsat	21 745	14	578	4	456
	gurobi	21 657	14	242	7	340
I.4_5_4_120	cpsat	28 263	17	710	3	636
	gurobi	28 276	17	470	3	509
I.8_10_2_120	cpsat	63 473	44	1757	2	1 243
	gurobi	63 628	44	614	2	867
I.8_10_3_60	cpsat	32 923	24	922	2	1 426
	gurobi	33 032	24	819	2	1 539
I.8_5_2_90	cpsat	20 841	14	956	3	643
	gurobi	21 296	14	385	5	464
I.8_5_4_60	cpsat	18 836	13	314	2	762
	gurobi	18 744	13	603	2	622
mean	cpsat		22.1	835	2.3	832
mean	gurobi		22.1	462	3.1	761

Table 9: **Pending: the case study has not been run against the current code.** The company baseline is rebuilt and this table is regenerated when that stage completes.

[case study pending]

Table 10: Measured solve time, compact 3D-SCPTOP against MATH-3D-SCPTOP, under an identical 1800-second budget for each method. $\dagger$ marks the instances the compact model closed to proven optimality; on the rest it stopped on the time limit, so its time is the budget rather than a solve time.

instance	compact (s)	MATH (s)	MATH/compact	Δz
I.4_10_2_90	1800	1194	0.66	—
I.4_5_2_60	1800	959	0.53	-0.20%
I.4_5_3_90	1800	797	0.44	+0.84%
I.8_5_2_90	1800	959	0.53	+10.83%
S.2_2_2	1800	624	0.35	+27.99%
S.2_2_3	1800	622	0.35	+0.00%
S.2_3_2	1800	855	0.48	-0.02%
S.2_3_3	1800	417	0.23	+0.00%
S.3_2_2	1800	663	0.37	+0.21%
S.3_2_3 †	11	616	55.50	+0.03%
S.3_3_2	1800	901	0.50	-0.10%
S.3_3_3 †	14	424	29.82	+0.06%

Table 11: Measured solve time by approach. Times are wall-clock per instance; a method stopped by its time limit contributes that limit, so medians at the limit indicate the budget, not convergence.

approach	set	n	median (s)	min	max
compact 3D-SCPTOP	12 small instances	12	1800	11	1800
MATH-3D-SCPTOP	same 12 instances	12	730	417	1194
MATH-3D-SCPTOP	benchmark instances	5	1745	1545	2020
SCPTOP (volumetric)	benchmark instances	5	0	0	1
pricing oracle: cpsat	oracle stage	9	762	456	1426
pricing oracle: gurobi	oracle stage	9	710	340	1539

falls to 54% and 7.8%. Coarse containers leave large unusable remainders once footprints and layer heights are respected, and it is exactly that remainder a volumetric model spends. The direction of the error is worth stating plainly: a volumetric capacity is not a safe approximation that merely loses a little precision — it is systematically optimistic, and most so when the containers are large, which is the regime of interest in automotive procurement.

Feasibility by construction. Every one of the 36 plans MATH-3D-SCPTOP returns passes an independent truck-by-truck verification against (38)–(43), performed by code that shares nothing with the solver that produced it. That is not a coincidence of the search but a property of the formulation: a column is a witnessed layout, so a plan assembled from columns is executable by construction, and can be handed to a loading crew as a drawing. The realised utilisation confirms the loads are not merely legal but full: the least-filled truck in a plan reaches 68% of the binding resource on average, never below the $\alpha = 0.55$ floor.

Against the procedure in use. On the case study built from the industrial catalogue, MATH-3D-SCPTOP reduces total cost by 11.4–12.8% and the number of truck dispatches by 17–18% relative to the company’s average-mix procedure, consistently across the low-, normal- and high-variability scenarios. Against the stronger hand-drawn baseline — regular grids plus two-zone layouts, which is more than the planner actually maintains — the reduction is 10.5–11.9%. Two further observations matter more than the percentages. First, the advantage does not erode as demand variability grows; if anything it widens slightly, because a fixed set of average-mix configurations tracks a varying programme less and less well. Second, the size of the configuration set matters as much as its content. Reconstructed from average mixes alone the set is very small when demand is stable — at low variability the horizon average and the per-period mixes coincide — and too small to serve the programme. That is a statement about the reconstruction rather than about the procedure: the plant operates, so the set its crews work with is feasible by construction. Rebuilding the variety they accumulate in practice — one pattern per dominant product, single-product patterns, several fill levels — restores feasibility, and the number of configurations needed to do so is itself the finding: on the low-variability instances an averaging rule supplies fewer than sixty where serving the programme takes around two hundred. Keeping a fixed configuration set feasible as the programme moves means maintaining a large catalogue by hand, every entry of it found by trial.

Pricing technology. The subproblem was solved both as a constraint programme and as a MILP with the big- M non-overlap encoding (16)–(19), under identical budgets, by running the whole matheuristic with each backend on the same nine instances (Table 8). Neither technology dominates. The mean ratio of objectives is 0.9997 — a tie — and the per-instance scatter runs from -6.4% to $+4.6\%$, so the difference between them is smaller than the variability of either. Both return feasible plans on every instance.

What does differ is how they get there. The MILP reaches comparable plans with roughly half as many columns (459 against 875 on average) and slightly less time (768 s against 866 s), and it certified a bound on one instance where the constraint programme certified none. The interpretation we prefer is that the disjunctive structure of non-overlap is propagated more cheaply by a global constraint, so the CP oracle answers each pricing call faster and the column generation therefore takes more, smaller steps; the MILP answers fewer calls but each is more informative. Neither effect is large enough to matter for the plans that come out.

We report this because the natural expectation — that a packing subproblem must favour constraint programming — is not what the measurements show, and because an earlier version of this comparison appeared to show a clear CP advantage. That version was invalid: the MILP oracle then imposed the

full-truckload requirement as the sum of the volume and weight ratios rather than as the disjunction (42), so it was solving a strictly weaker problem and one of its columns failed verification. Once both oracles solve the same problem the advantage disappears. We use CP-SAT for the large instances on the strength of the marginally better mean, but the choice is immaterial.

On bounds, and what the method does not deliver. The lower bounds are the weak point of these results and we report them as such. Certification requires the packing subproblem to be closed to proven optimality at the final pricing sweep, which succeeds on only 3 of the 36 instances; elsewhere the Lagrangian bound (50) applies, and it is loose — a mean gap of 28.6%, which reflects the bound rather than the plans. Where the compact model can be solved, the true gaps are far smaller (Table 7) — a median of about 2%, with two outliers on the two-product instances where the plan uses only a couple of trucks and the objective is dominated by inventory cost, so that a single over-delivered truckload moves it appreciably. That is the honest way to read the large-instance gaps: as an artefact of the bounding argument rather than a measure of the cost being left on the table. Closing them would require embedding the column generation in a branch-and-bound tree rather than the depth-first dive used here, which is the natural next step and is beyond the scope of this paper.

Where each method belongs. Table 7 sets the compact model against the matheuristic on every instance where the former produces anything, warm-started with the latter’s plan so that it begins from a feasible incumbent. Both methods receive the same wall-clock budget, which earlier revisions of this experiment did not: the matheuristic was given 400 seconds against the compact model’s 1800, and the deficits reported at that setting were an artefact of the handicap rather than a property of the method. Under equal budgets the matheuristic matches or improves on the compact model on five of the twelve instances — twice to the cent, and by 0.02 to 0.20% on three more — and lies within 0.9% on four others, using between a quarter and two thirds of the wall-clock. Where the compact model closes an instance outright it remains the better tool: it proves optimality in eleven and fourteen seconds on the two three-product instances, against which the matheuristic gives away 0.03 and 0.06%. At ten periods the compact model returns nothing at all within half an hour, while the matheuristic returns a plan with a *certified* gap of 1.42%.

The honest reading is that the crossover is the planning horizon rather than the raw size of the formulation, and that for short horizons the compact model should simply be used. The case for the decomposition rests on the regime beyond it, where the compact model produces no plan and the matheuristic returns a verified one in twenty minutes; the industrial instances of Section 7 sit there.

One deficit is worth naming precisely, because it is neither a bounding artefact nor noise: on the eight-product instance the matheuristic dispatches fourteen trucks against the compact model’s twelve, a 10.8% difference that two independently derived lower bounds corroborate — the compact model’s own bound and the matheuristic’s Lagrangian bound agree to within 0.02%, and the incumbent lies above both. The remaining large percentage in the table is of a different kind: on a two-truck instance a single extra dispatch is 28% of the objective, where the same physical error costs 1.6 to 2.3% on the benchmark instances, and that instance converges to within 0.008% of the compact model once the pricing sweeps are given the threads and the time they need. Percentages on plans this small measure granularity rather than quality. The cause is granularity. The master can only ship integer combinations of the loads that have been generated, and when no such combination reaches the delivery target with the smaller fleet it must add a truck, whereas the compact model chooses each truck’s contents freely. Enriching the pool with reduced loads narrows this but does not close it. Generating loads against the residual requirement of a fractional master solution, rather than against dual prices alone, is the obvious remedy and is left for future work.

7 Practical application

This section presents a practical procurement transport application. The proposed resolution framework was evaluated using real data from an automotive industry supply chain. We validate and compare the 3D-SCPTOP solution procedures to the company’s procedure, which employs predefined loading configurations that indicate the number of each container type to be carried. Finding these loading configurations is a very tedious and time-consuming manual task that is based on a spreadsheet procedure. A mathematical programming model, dubbed as PLC-SCPTOP, which replicates the current’s company procedure, is then utilised to minimise the total cost, with a summation of procurement transport and inventory holding costs, and one based on these predetermined configurations.

The real-world data derived from the company’s operations act as the empirical foundation for the comparison. To broaden its scope we scale and perturb these data under a stochastic process, giving three scenarios that differ only in demand variability: low ($\pm 5\%$), normal ($\pm 15\%$) and high ($\pm 35\%$) around the same mean. Three procedures are compared on the same instances.

- **Company procedure.** PLC-SCPTOP over the configurations the planner derives today: a truck laid out to follow the *average* container mix across plants and periods, recomputed as the programme is revised, together with one standard single-container-type pattern per RTI type and the partial loads a planner will dispatch rather than miss a delivery.
- **Hand-drawn benchmark.** PLC-SCPTOP over a richer predefined set: a regular grid per container type plus two-zone layouts in which the floor is cut once and each zone tiled with a single footprint. This is more than the planner actually maintains, and is therefore the harder baseline.
- **MATH-3D-SCPTOP** of Section 5, which generates its configurations and augments the planner’s existing ones rather than replacing them.

Table 9 reports the outcome. MATH-3D-SCPTOP reduces total cost by 11.4–12.8% and the number of truck dispatches by 17–18% relative to the company procedure, and by 10.5–11.9% relative to the hand-drawn benchmark. Two features of the result matter more than the percentages.

First, the advantage does not erode as demand variability grows; if anything it widens slightly, from 11.4% under low variability to 12.8% under high. This is the expected behaviour of a fixed configuration set: an average mix tracks a varying programme progressively worse, and the gap it leaves is exactly what generated configurations recover. It also answers the concern that such a method should help only when configurations and demand are misaligned — the misalignment is not an occasional pathology but the normal state of a programme that varies at all.

Second, the size of the configuration set matters as much as its content. Reconstructing the company’s set from average mixes alone yields very few distinct configurations when demand is stable — at low variability the horizon average, the thirds and the per-period mixes all coincide — and a set that small cannot serve the programme. Crews do not work that way: they accumulate patterns by trial, one for the days a single product runs heavy, another for when two share the load. Reconstructing that variety (the average mix, one pattern per dominant product, single-product patterns, each at several fill levels) restores feasibility, and the number of configurations required to reach it is itself informative: on the low-variability instances the averaging rule supplies fewer than sixty, while serving the programme takes around two hundred.

That is the practical cost of a fixed configuration set. It is not that the procedure fails — the plant plainly operates — but that keeping it feasible as the programme moves requires the planner to maintain a large and growing catalogue by hand, and every configuration in it was found by trial rather than by design.

The managerial reading is that the benefit is not a better spreadsheet. It is that the loading configuration stops being an input fixed months in advance and becomes a decision taken with the transport plan, so that when the container mix moves the configurations move with it.

Therefore, this method shows promising results for reducing inventory and transport costs, particularly under variable demand and misalignment between loading configurations and container types.

8 Conclusions

We herein propose a MILP model, 3D-SCPTOP, to tackle procurement transportation planning at an operational level by incorporating stack-based full-truck loading constraints into its considerations. Given the NP-hard nature of the placement subproblem and the fact that the compact model replicates it once per loading configuration, we reformulate the problem by Dantzig–Wolfe decomposition, taking the complete loaded truck as the unit of decision, and solve it by branch and price with a constraint programming pricing oracle, giving MATH-3D-SCPTOP. Because every column is a fully specified truck layout, the plans the method returns are executable by construction and can be handed to a loading crew as a drawing; we verify each dispatched truck independently of the solver that produced it.

A case study has also been conducted within an automotive SC that comprises a car assembler, a first-tier supplier and a second-tier supplier. The objective of the 3D-SCPTOP model is twofold: (i) to minimise the total number of trucks used; (ii) to minimise inventory holding costs. This approach takes into account the complexities of supply chain logistics in the automotive industry by demonstrating a successful integration of procurement transport planning and 3D loading constraints.

Managerial implications in a SC context are oriented to the proposed solution methodology, improving the solutions found by the company’s procedure. It is also able to adapt to demand changes while finding high-quality solutions in reasonable computational times.

A second managerial finding concerns the representation of transport capacity. Procurement transport models in the literature typically express it volumetrically. Our comparison shows that doing so is not a harmless simplification: the volumetric model returns plans that are cheaper on paper precisely because most of the truck loads it prescribes cannot be placed in a trailer at all. A planner acting on such a plan discovers the shortfall on the loading dock, not in the model.

Nevertheless, several challenges remain for further research lines to deal with; for instance, the inclusion of multiple suppliers to better represent complex supply chains. A more comprehensive transport operation model could be achieved by incorporating routing aspects, such as vehicle routing with 3D loading constraints. Future work may also relax the layer-homogeneity assumption to allow intra-layer mixing of RTI types, or extend the placement block with axle load-balance constraints. On the algorithmic side, the search already keeps both children of a branch and re-prices at every node, but reaching integrality by branching alone remains out of reach within a practical budget, and the incumbent still comes from an integer master solved over the pool; closing that gap, together with certifying the bound at every node rather than once at the end, would turn the matheuristic into an exact method. Integrating sourcing and delivery transport decisions could improve supply chain efficiency. Finally, the incorporation of production scheduling into the model has the potential of offering a cohesive approach to supply chain planning optimisation by aligning production timelines with transportation and inventory strategies.

A The SCPTOP benchmark

The benchmark against which 3D-SCPTOP is compared is an extension of the procurement transport model of (Díaz-Madroño, Mula, and Peidro 2014) to the closed-loop, two-node setting studied here. Three modifications are made to that model, as stated in Section 6: fuzzy parameters are dropped; the transport capacity restriction is expressed through the volume of the individual RTIs against the truck volume rather than a maximum container count; and a product may travel in more than one RTI type.

The point of the comparison is to isolate the effect of the loading representation, so SCPTOP is made *identical to the transport and inventory block of 3D-SCPTOP in every other respect*: the same two nodes, the same closed loop of full and folded RTIs, the same lead time, the same coverage stock, the same payload cap, the same full-truckload floor and the same objective. It differs in exactly one place — where 3D-SCPTOP asks whether the RTIs can be stacked and placed on the trailer floor, SCPTOP asks only whether their volumes sum to less than the trailer’s.

Let $Y_{gkt} \in \{0, 1\}$ indicate that truck k is used in direction g in period t , and let Q_{nt}^k and Q_{rt}^k be the full RTIs of product n , respectively the folded RTIs of type r , that it carries. All remaining notation is that of Table 3.

$$\text{Min } z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} cK Y_{gkt} + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} ic_{gn} I_{gnt} \quad (70)$$

subject to, for all $k \in K$ and $t \in T$,

$$\sum_{n \in N} \gamma_{pp(n)} Q_{nt}^k \leq \Gamma_K Y_{pkt}, \quad \sum_{r \in R} \gamma_{sr} Q_{rt}^k \leq \Gamma_K Y_{skt} \quad (71)$$

$$\sum_{n \in N} w_n Q_{nt}^k \leq pK^w Y_{pkt}, \quad \sum_{r \in R} w_r^f Q_{rt}^k \leq pK^w Y_{skt} \quad (72)$$

together with the full-truckload floor on the binding resource, imposed exactly as in (42): a dispatched truck must reach a fraction α of *either* the trailer volume or the payload,

$$\sum_n \gamma_{pp(n)} Q_{nt}^k \geq \alpha \Gamma_K (Y_{pkt} - 1 + \theta_{kt}), \quad \sum_n w_n Q_{nt}^k \geq \alpha pK^w (Y_{pkt} - \theta_{kt}), \quad (73)$$

with $\theta_{kt} \in \{0, 1\}$ and the supplier direction analogous. The flows aggregate as $Q_{nt} = \sum_k Q_{nt}^k$ and $Q_{rt} = \sum_k Q_{rt}^k$, and the inventory balances, coverage stock and domains are those of (2)–(12), unchanged.

Constraint (71) is the whole of the benchmark’s loading model: a truck is deemed able to carry any set of RTIs whose volumes fit inside the trailer’s. No footprint, stack height, layer count or crush limit appears. Section 6 reports what that omission costs — not in solution quality, where SCPTOP appears cheaper, but in whether the plans it prescribes can be loaded at all.

Why the floor is imposed on the binding resource. Writing (73) on volume alone, as the original formulation does, makes dense freight structurally infeasible: filling α of a 97.9 m³ trailer with RTIs at 590 kg/m³ weighs 27.4 t against a 24 t cap, so no truck is dispatchable and the instance is empty. Summing the two ratios instead is strictly weaker — it accepts a truck 35% full by volume and 20% by weight. The disjunctive form above is used identically in every model compared here, so no method is held to an easier standard than another.

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